

Chapter 48. Monitoring during Mechanical Ventilation

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Monitoring during Mechanical Ventilation: Introduction

Patients are admitted to an intensive care unit (ICU) for two main reasons. One is for delivery of mechanical ventilation, the varied aspects of which are the subject of this book. The second is to observe a patient more closely than is possible on a hospital ward; that is, to avail of specialized devices used for the monitoring of vital functions (and of staff who have expertise in their operation).

The literature published on these two subjects constitutes the unique corpus of knowledge required for the expert practice of intensive care medicine. Management of some critically ill patients is based on knowledge outside these two areas, such as patients with acute gastrointestinal bleeding or acute renal failure. The principles for managing such conditions have been developed by physician-investigators within the relevant subspecialties rather than by intensive care physicians. Beyond the areas of mechanical ventilation and monitoring, an intensivist seeking the most authoritative writing on a subject must turn to articles and texts published by nonintensivists.

Patients receiving mechanical ventilation are exposed to the full range of monitoring devices. An in-depth discussion of each device would require as much space as the rest of this book. For such discussions the reader is referred to the companion text, *Principles and Practice of Intensive Care Monitoring*.¹ Rather than attempting a synopsis of each individual monitoring technique, this chapter provides a bird's-eye view of the subject. We map out the territory of monitoring through discussion of its goals, the principles of measurement, usefulness of monitoring in various settings, the forms of clinical reasoning used in interpreting generated data, evaluation of the benefit of monitors, and the problems in designing studies that attempt such evaluations. To give a sense of the detailed contours of the terrain being traversed, we offer specific examples of how the general topics relate to the everyday use of monitors.

What Is Monitoring?

An unsatisfactory—and embarrassing—aspect of writing about monitoring is the absence of a generally accepted definition. A definition tries to set criteria that demarcate the boundaries between concepts (or things) so as to prevent overlapping or confusion, and has the goal of providing order or a clear understanding.

The word *monitor* comes from the Latin *monere*, which means “to warn.” This meaning connotes one goal of monitoring, to provide an alert, an alarm. But monitors serve other functions. Monitors replicate many of the characteristics of diagnostic testing, and interpretation of the generated data must comply with the scientific principles developed for use of diagnostic tests. Thus, it might seem reasonable to define monitoring as the serial performance of diagnostic tests at frequent intervals. But how should we demarcate the frequency at which measurements are repeated? Certain signals have such a high-frequency content that obtaining fifty measurements in a second is insufficient to capture a critical data point.

Having struggled long and hard in trying to formulate a meaningful but precise definition of monitoring, we admit defeat. We can do no better than offer a naturalist's definition: Monitoring refers to what physicians and allied health personnel do with monitors.

Goals of Monitoring

The major goals of monitoring are to continuously measure key variables that enhance understanding of a patient's underlying disease state, to aid with diagnosis, and to guide management.¹ To accomplish these goals, the monitor should measure a variable that is pertinent to the disease process, be technically accurate, provide interpretable data, be practical for use, and not cause harm.^{1,2} The ability of today's monitors to meet these goals depends largely on two factors: whether a monitor is capable of accurately measuring the function or disease manifestation in question, and whether the recorded information will help improve patient outcome. Inextricably linked to these two factors is the caregiver's ability to interpret data, integrate data from multiple sources, discard data that are not important, and then make a wise judgment about diagnosis and therapy.

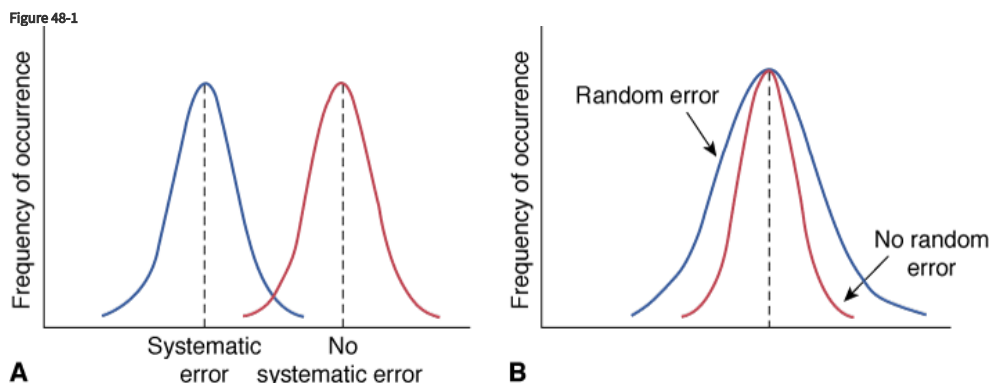
Accuracy of Measurements

Advances in electronic and computer technology over the last 30 years have generated several monitoring systems that are accurate and easy to use. To know how to operate a machine and interpret the provided data, a physician needs much more than a manufacturer's manual. The physician needs to know the principles of the monitoring device, how to operate it, the reliability of the measurements, indications and contraindications, complications with use of the device, and how to troubleshoot problems as they arise.

To determine whether the data generated by a monitor are valid, a physician needs to have a basic understanding of measurement theory and its applications. For an extensive review of the science of measurement, the reader is referred to the chapter by Chatburn³ in the companion book, *Principles and Practice of Intensive Care Monitoring*.⁴

Fundamentals of Measurement Theory

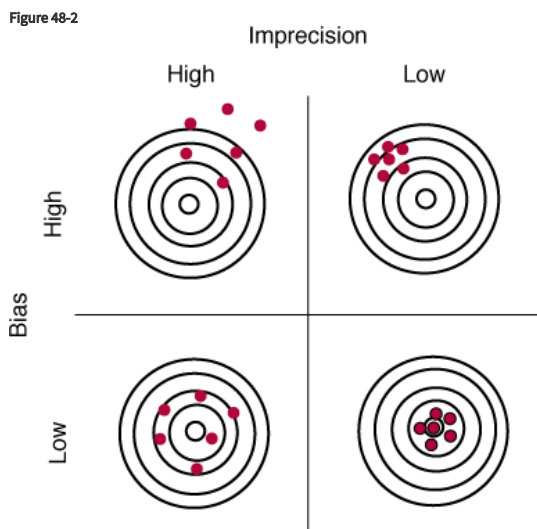
Every measurement has errors. Accordingly, the observed measurement is a mixture of the true value and the error. Errors can be systematic or random. Systematic error (also termed bias) is the difference between the mean value of repeated measurements and the true value. Systematic errors occur in a predictable manner; they cause measurements to *either* consistently underestimate or consistently overestimate the true value (Fig. 48-1A). These type of errors are not affected by repeated measurements but can be reduced by proper calibration.



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Distribution of measured values in the presence and absence of systematic (A) and random (B) errors. A systemic error increases or decreases the average value. A random error increases the variability around the average value.

Random error (a measure of the imprecision of a measurement) is the difference between the mean of the sample and the measurement of individual values. Random errors occur in an unpredictable manner because of uncontrollable factors; they cause measurements to *both* underestimate and overestimate the true value (see Fig. 48-1B). Increasing the number of measurements can reduce random errors.³ Figure 48-2 illustrates the effects of bias and imprecision (systematic and random errors) on measurements.

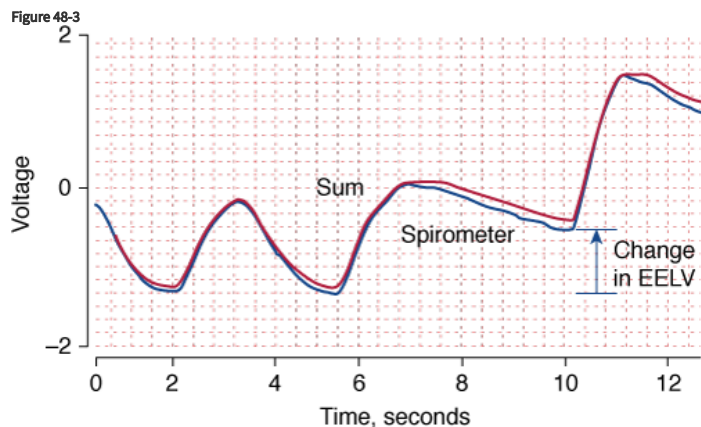


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The effects of bias and imprecision (systematic and random errors) on measurements using the analogy of target practice at a rifle range. When bias is low, measurements (red circles) group around the true value (bull's-eye). When imprecision is low, the cluster of measurements is tight and the random error of repeated measurements (rifle shots) is small. Ideally both bias and imprecision are low, resulting in a small total error for repeated measurements. (Used, with permission, from Chatburn.³)

Accuracy

Accuracy is the ability of a measuring device (or monitoring technique) to capture the true value of a quantity. When assessing accuracy, researchers commonly compare a new method with an established method to determine whether the new method can replicate the established method (Fig. 48-3).



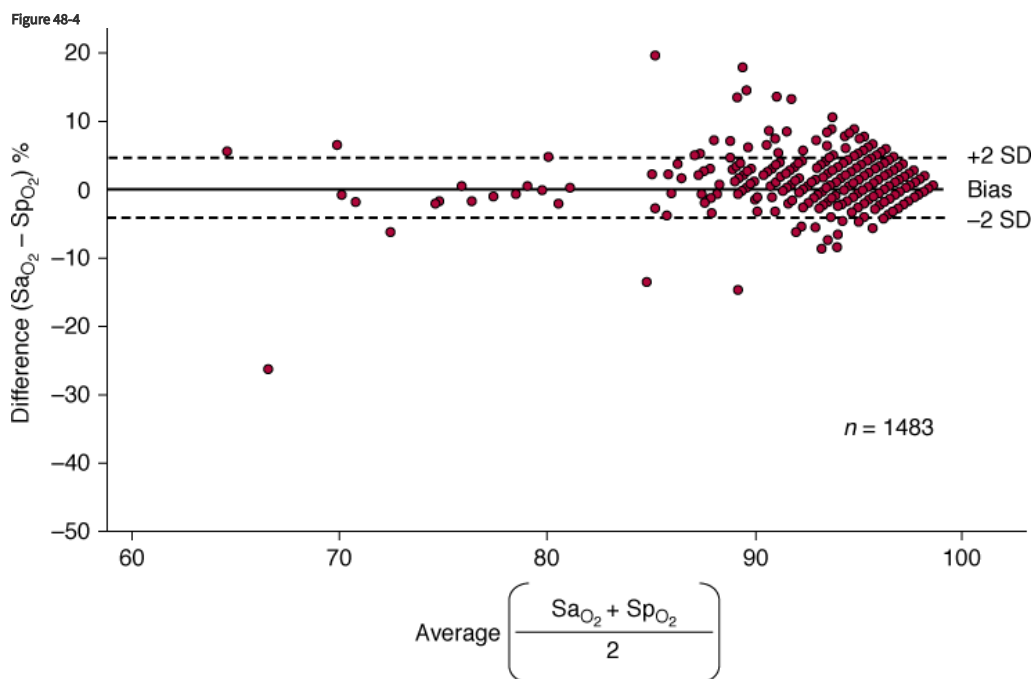
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Changes in end-expiratory lung volume measured simultaneously with a calibrated respiratory inductive plethysmograph (*red*) and a spirometer (*blue*), the reference instrument. The two signals are almost superimposed upon one another indicating good agreement between the two techniques. *EELV*, end-expiratory lung volume.

Accuracy is usually expressed as a percentage of the full scale. In today's ventilators, pressure-measuring systems have an accuracy of ± 2 cm H₂O plus 4% of the signal. In other words, if the true pressure is 25 cm H₂O, the displayed pressure could be anything between 22 and 28 cm H₂O.⁵ In contrast, the accuracy of flow sensors within ventilators varies much more—from $\pm 1\%$ to $\pm 20\%$ of the setting, depending on the ventilator.⁶ Despite their common use, remarkably little information on the accuracy of flow-measuring and pressure-measuring devices has been published in peer-reviewed journals; in general, users must resort to data on accuracy provided by the manufacturers.

The monitoring device that has been studied the most in critically ill patients is the pulse oximeter. Compared with the measurement standard (a multiwavelength Co-oximeter), pulse oximeters have a mean difference (bias) of less than 1% and a standard deviation (precision) of less than 2% when arterial oxygen saturation (Sa_{O_2}) is 90% or above (Fig. 48-4).^{6–8}



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Differences between arterial oxygen saturation (Sa_{O_2}) measured by a blood-gas analyzer and oxygen saturation measured by pulse oximetry (Sp_{O_2}) plotted against the average O₂ saturation in 102 critically ill patients. The *solid line* is the bias and the *dashed lines* are 95% confidence intervals. Mean difference (bias) between two measurements is small. *SD*, standard deviation. (Reproduced, with permission, from Van de Louw A, Cracco C, Cerf C, Harf A, Duvaldestin P, Lemaire F, et al. Accuracy of pulse oximetry in the intensive care unit. *Intens Care Med*. 2001;27:1606–1613. With kind permission from Springer Science and Business Media.)

Precision

Precision is the ability of a monitor to display the same value repeatedly assuming that the actual quantity has not changed; that is, the degree of consistency between repeated results. Precision is usually expressed in terms of variance, standard deviation, or confidence intervals. In ventilated patients, the precision (quantified as standard deviation) of pulse oximetry in measuring oxygen saturation, SpO_2 is less than 1.2% when SpO_2 is greater than 90%, and it increases to 2.7% when SpO_2 drops to 90% or less.^{6–8}

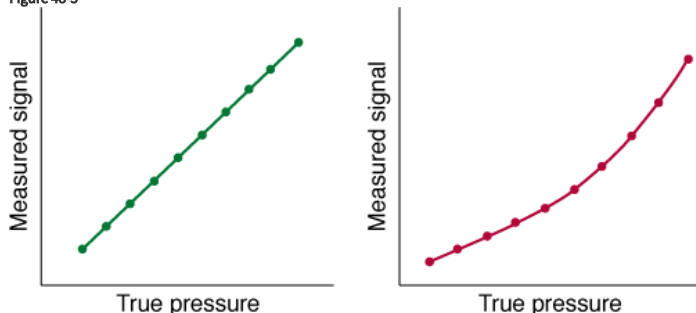
Reproducibility is the ability of the monitoring technique (or device) to maintain its precision during long-term use. Knowing the reproducibility of a measurement is important for judging whether a change is truly of clinical significance. The reproducibility of the thermodilution technique in measuring cardiac output, quantified as the ratio of standard error to average cardiac output, varied between 2% and 5% when measurements were repeated (an average of three times), and varied between 2.5% and 8.7% for single measurements.⁹ Thus, before concluding that a change in cardiac output is clinically significant, a clinician would need to observe a 12% to 15% difference between repeated determinations (three measurements per determination).

A technique may yield reproducible measurements at one sitting, but be unreliable when employed by a different investigator or on a different day. In ventilated patients, a single investigator obtained highly reproducible measurement of maximal inspiratory pressure at a single sitting: The coefficient of variation for triplicate efforts was 11%.¹⁰ When different investigators undertook the measurements, however, the coefficient of variation was $32\% \pm 4\%$, despite the measurements being obtained in the same patient and on the same day.

Linearity

A device is considered linear when a plot of output data from the device (the measured values) versus the input data (the known values) can be fitted with a straight line. Occasionally, a system measures pressure accurately only over part rather than the entire range of pressures; that is, the system is **alinear** (Fig. 48-5). Alinearity may occur because the given transducer is not designed for use over the entire range or because it is not functioning properly.

Figure 48-5



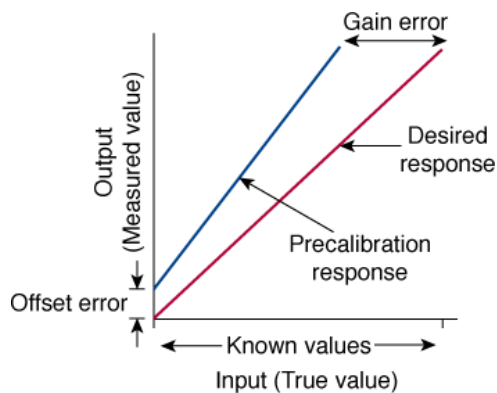
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A measured pressure signal plotted against a reference measurement of pressure ("true pressure") in a linear (*left panel*) and nonlinear (*right panel*) pressure-measuring system. (Used, with permission, from Gallagher.⁵)

Calibration

Calibration is the process of adjusting the output of a device to match a known input value. Thus, systematic error is minimized. Calibration is considered static when pressure is changed in steps and each step is maintained long enough to achieve a stable signal output. For a linear measurement system, calibration is a simple two-step procedure (Fig. 48-6). First, the readout is set at zero while no input signal is applied to the instrument. If an offset error occurs, it is adjusted. For a flow-sensing device, the flow sensor is occluded and a reading of no flow should result. If some flow is registered, the readout is adjusted to zero to correct for the offset error. Next, sensitivity (gain or slope) is set by applying an input signal of known value (say 12 L/s from a rotameter), preferably at the upper end of the output range; the readout on the flow sensor is then adjusted to read 12. If the system is linear, the readouts for all input values between these two calibration points will be accurate.

Figure 48-6



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The two-point calibration procedure. First, the readout on the instrument is adjusted to read zero to overcome offset error. Then, the gain is adjusted to read the desired response. See text for details. (Used, with permission, from Chatburn.³)

The dynamic response of a measurement system, such as a pressure transducer, is typically tested using a square wave (or sinusoidal) pressure generator. A step change in pressure is followed by a brief delay before the measured pressure begins to change.¹¹⁻¹⁴ The amplitude of the test should be within $\pm 5\%$ of the amplitude of the reference system at the highest frequency tested. The phase lag is the temporal difference between the two systems.

Sources of Measurement Error during Monitoring

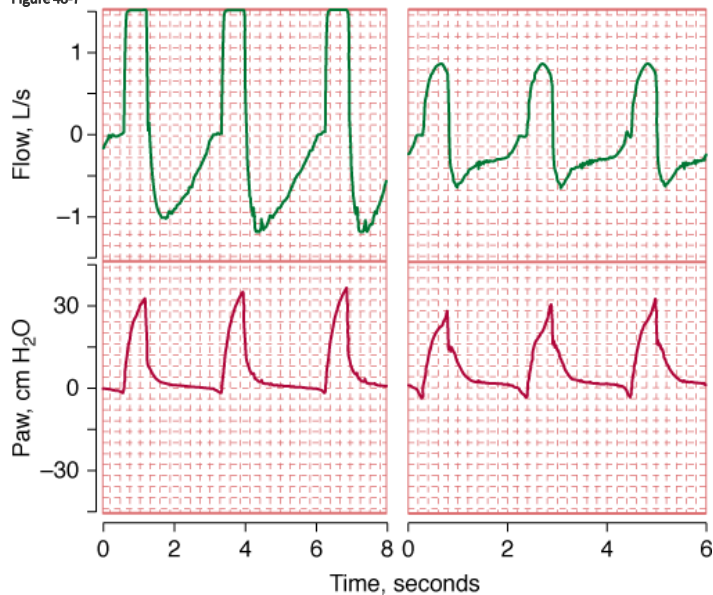
A major challenge in interpreting data from monitors is to determine which data constitute truth and which constitute artifact. Although both systematic and random errors can cause artifact, the major source of error in biologic research is systematic errors.¹⁵

Systematic Error

Zero offset error occurs when the zero point is not correctly set during calibration but the gain is correct. Consequently, the instrument will read consistently either an inaccurately low or an inaccurately high value over the entire scale. Drift error is a form of time-dependent offset error in which the changes occur over time. The zero point can also change secondary to patient movement without staff being aware.

Range errors occur when the true value of the input system is beyond the operating range of the device. Signals that are either below or above the calibrated scale value may be clipped (the displayed value is different from the true value) (Fig. 48-7).

Figure 48-7



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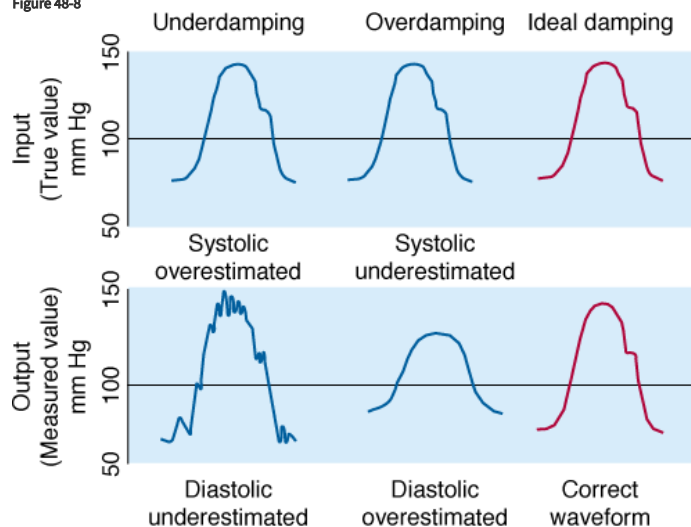
Flow and airway pressure in two patients receiving assist-control ventilation at a flow rate of 60 L/min. For the patient in the *left panel*, the flow signal is “pinned” or “clipped.” The clipping of the flow signal in the left panel was caused by clogging of the pneumotachograph with secretions.

Response time is the time that a monitoring device takes to respond to a step change in a recorded physiologic variable. The 90% response time is the time from the step change in pressure until the measured signal settles within 10% of its final value. The response time for a pulse oximeter probe varies with its

location. Probes placed on the ear respond more rapidly than probes on the finger.^{16–18} During calibration of a monitoring device, a slow response time causes error if the user does not allow sufficient time for the instrument to stabilize at the known value.

Frequency response is an instrument's ability to accurately measure an oscillating signal. When the frequency of a signal increases, a measurement system will generate either an underestimate (attenuation) or overestimate (amplification) of the true amplitude of the signal. An optimally damped system will measure all signal frequencies within the working range with equal amplitude (Fig. 48-8).

Figure 48-8



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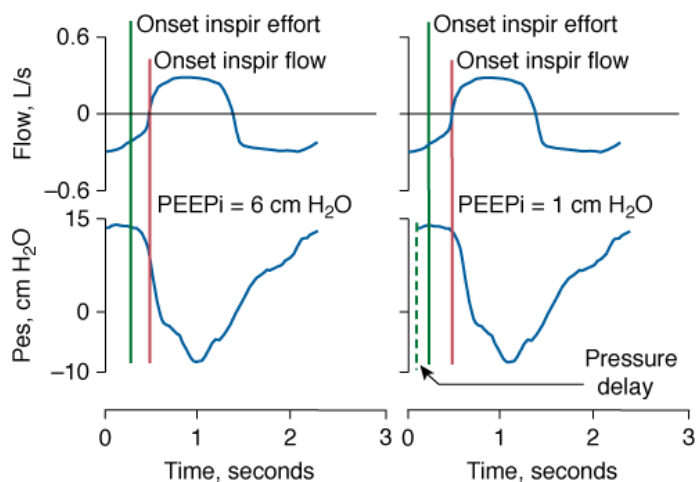
Effect of three different blood pressure transducers, which exhibited different damping characteristics, on the measurement of blood pressure. (Used, with permission, from Chatburn.³)

A system is considered damped when some of the signal-component frequencies are attenuated. Overdamping causes systolic pressure to decrease and diastolic pressure to increase. In a system for measuring vascular pressure, sources of overdamping include air bubbles in the transducer or pressure tubing, clot or fibrin at the catheter tip, and a kinked or partially occluded catheter. Underdamping amplifies the higher harmonics, producing oscillations, which obscure systolic and diastolic values. As such, underdamping causes systolic pressure to increase and diastolic pressure to decrease. Common sources of underdamping include an increase in heart rate interfering with the natural frequency of the pressure-transducer system; a “catheter whip,” commonly observed with a pulmonary artery (PA) catheter, can produce an underdamped signal (see Fig. 48-8).

PA waveforms generally contain more high-frequency information than do systemic arterial waveforms, and small errors in pressure assume more clinical significance. Poor dynamic frequency response and overdamped pressure tracings accounted for over half the technical problems encountered in measuring PA wedge pressure.¹⁹ A simple test for assessing the dynamic response of a vascular pressure catheter system is the rapid-flush test.²⁰

Alignment of Signals. When employing two different signals to quantify lung mechanics (flow and pressure), it is important to ensure the absence of a phase lag between the two signals. A lag can induce large errors in measurement of resistance and compliance.⁵ Phase lags also cause major errors when dynamic intrinsic positive end-expiratory pressure (PEEPi) is measured using an esophageal balloon-catheter system. If a phase lag causes the esophageal signal to occur before flow, PEEPi will be overestimated or considered present when it is truly absent. If a phase lag causes the esophageal signal to occur after flow, PEEPi may be underestimated (Fig. 48-9).

Figure 48-9



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Left panel. Flow and esophageal pressure (P_{es}) signals are properly aligned. Intrinsic PEEP ($PEEP_i$), measured as the difference between the onset of inspiratory (*Inspir*) effort and onset of inspiratory flow, is 6 cm H₂O. **Right panel.** Flow and P_{es} signals are not properly aligned in terms of phase: flow precedes P_{es} , causing underestimation of $PEEP_i$.

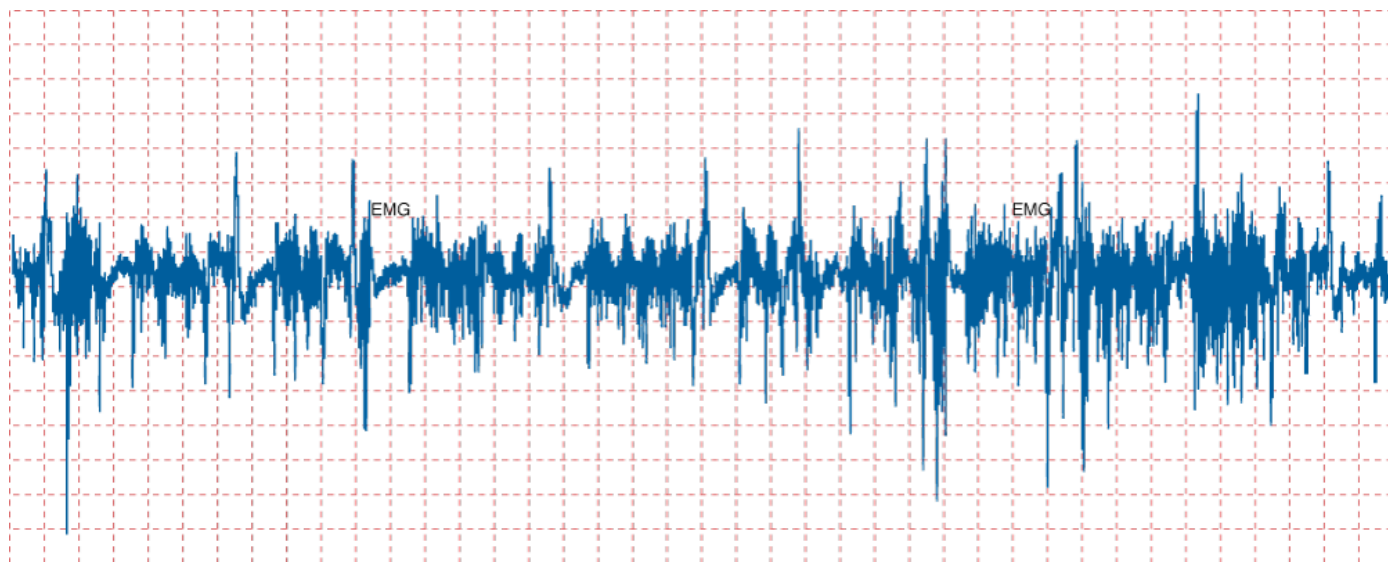
Variable Conditions between Calibration and Data Collection. If a measurement system is used under conditions that differ from the conditions of calibration, and no correction is made, systematic errors may result. Flow or volume readings on a ventilator are inaccurate if the temperature or humidity during the recording differs from those present during calibration.^{21,22} During calibration, volume may be measured under conditions of ambient temperature, with gas fully saturated with water vapor. If appropriate corrections are not made, subsequent readings of volume, under conditions of normal body temperature and with pressure saturated with water vapor, will underestimate the true volume of gas moving through the lungs. A mass of gas that is 1000 mL at 21°C (69.8°F) can become 1095 mL at 37°C (98.6°F).

Pneumotachographic measurements during mechanical ventilation will become inaccurate if the dimensions of the ventilator tubing during data recording differ from the dimensions used during calibration. The differences in air turbulence and the distribution of flow through the resistive element under the two conditions can lead to measurement errors. When a pneumotachograph was initially calibrated under ideal conditions (with tubing that provided optimized flow characteristics) and then subsequently attached to commonly used ventilator tubing, flow was underestimated by as much as 10% and volume by as much as 15%.²³

Random Error

Noise. All measurements are subject to some degree of rapidly changing interference, termed *noise* or *artifact* (Fig. 48-10).²⁴ The source of noise can be difficult to trace. Noise distortions occur randomly, so their effects are lessened if measurements are repeated sufficiently. The distortions are not reduced by calibration. The efficiency with which the signal can be distinguished from background noise is defined as the signal-to-noise ratio. Unless the ratio exceeds 1, the signal will be undetectable.

Figure 48-10

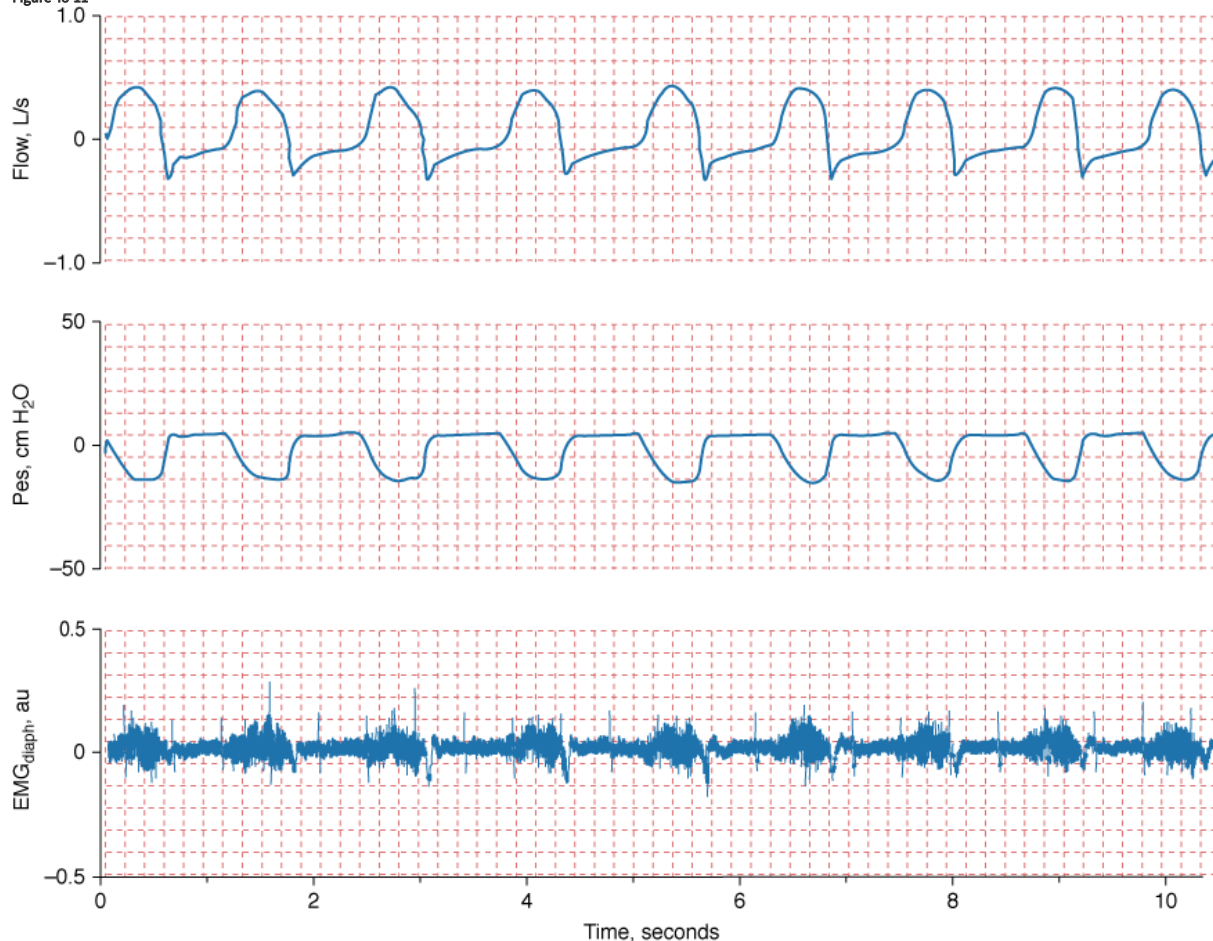


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Electromyogram tracing of the frontalis muscle reveals considerable noise.

Noise is particularly troublesome when weak signals are highly amplified. Noise is then amplified along with the signal, which eventually limits the sensitivity of the measurement. Cardiac contractions can distort an esophageal pressure signal such that pressure swings appear greater than they truly are. Likewise, the electromyogram signal can be contaminated by the electrical activity from other muscles (Fig. 48-11). As a signal is being transmitted to an amplifier, it is subjected to electrostatic and electromagnetic noise from nearby power lines (Fig. 48-12).²⁵ Radiofrequency noise may also be added from cellular phones. Finally, physical movement of the catheter cable changes its capacitance and may add low-frequency noise.

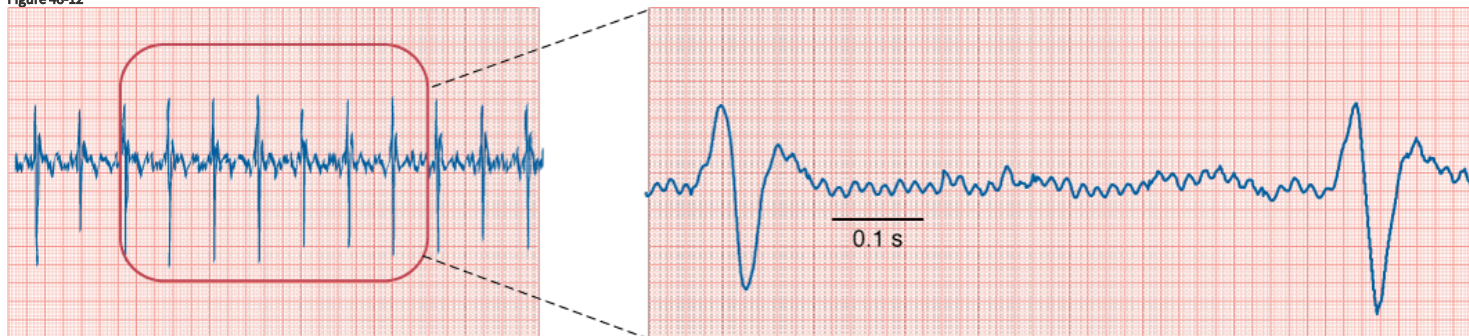
Figure 48-11



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Flow, esophageal pressure (P_{es}) and electromyogram of the diaphragm (EMG_{diaph}). The diaphragmatic EMG signal is contaminated by spikes from the electrocardiogram signal.

Figure 48-12

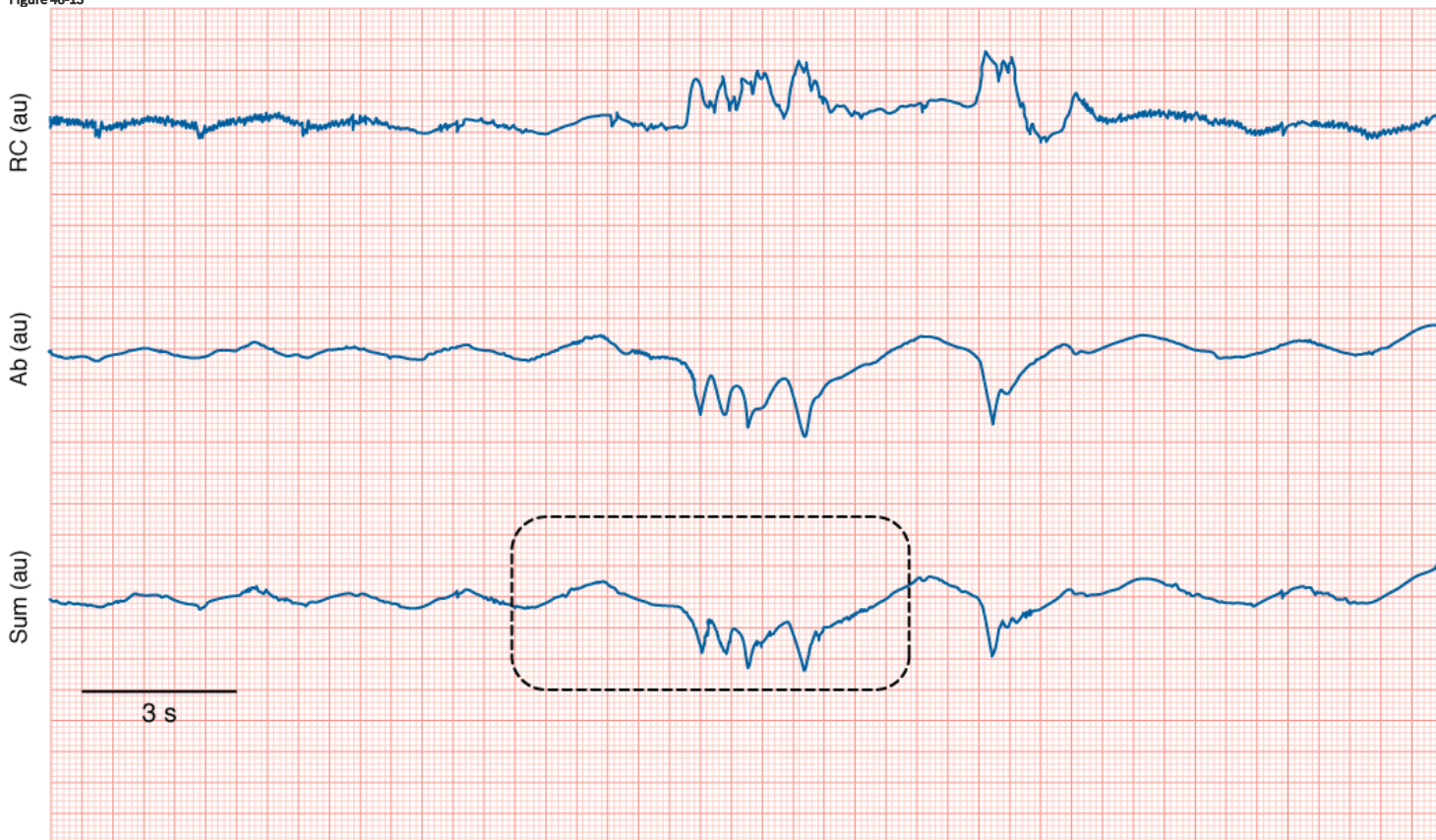


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Electrocardiogram tracing (*left panel*) exhibits background noise. On magnification, the noise in the signal is occurring at a rate of 60 cycles per second (a frequency of 60 Hertz), which is the frequency of electrostatic and electromagnetic signals of power lines.

Physiologic factors, such as a patient coughing, can lead to error or noise (Fig. 48-13). Physical factors such as slippage of transducers or improper placement of electrodes are also sources of noise. A pulse oximeter probe slipping off the patient's skin accounted for 35% of false alarms recorded in an ICU.²⁶ Patient movement disturbs the electrode-skin interface and causes electrocardiogram (ECG) noise.²⁷ "Ringing" after flushing an arterial catheter can interfere with the blood pressure signal and cause faulty readings.²⁸

Figure 48-13



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Recordings of motion of the rib cage (RC) and abdomen (Ab) obtained with a respiratory inductive plethysmograph in a patient being weaned from mechanical ventilation. The noise in the signal (*dotted rectangular area*) was caused by the patient starting to cough.

A particularly annoying consequence of signal noise is false alarms.^{29–31} Investigators prospectively waited at the bedside for 298 hours in an adult ICU. Of 2942 alarms, 2525 (86%) were false.²⁶ In a pediatric ICU, 68% of alarms were false.³⁰ A high rate of false alarms causes noise pollution, wastes nursing time, and contributes to staff burnout.³²

Nonlinearity. Nonlinearity introduces unpredictable error that varies over the operating range. With measurements of O_2 consumption, nonlinearity of the O_2 -sensing system causes a progressive increase in error as fractional inspired oxygen concentration (FI_{O_2}) is increased from 0.21 to 1.0.^{33,34} Thus, error in measurements of O_2 consumption increases from 2.6% at room air to 17% at an FI_{O_2} of 0.80.³³

Human Error. Human mistakes in acquiring data are another source of error. Depending on the angle at which the needle on an analog scale is read, different people can perceive different values for the same actual plateau pressure. Measurements that require patient cooperation introduce another source of error. Use of the end-expiratory airway occlusion method to measure PEEPi requires a cooperative patient who does not resist the occlusion.³⁵ Of 283 attempts in ventilated patients, PEEPi could be quantified in only eighty-six (30%); most of time PEEPi could not be quantified because the patient made an expiratory effort during the occlusion maneuvers (Fig. 48-14).³⁶



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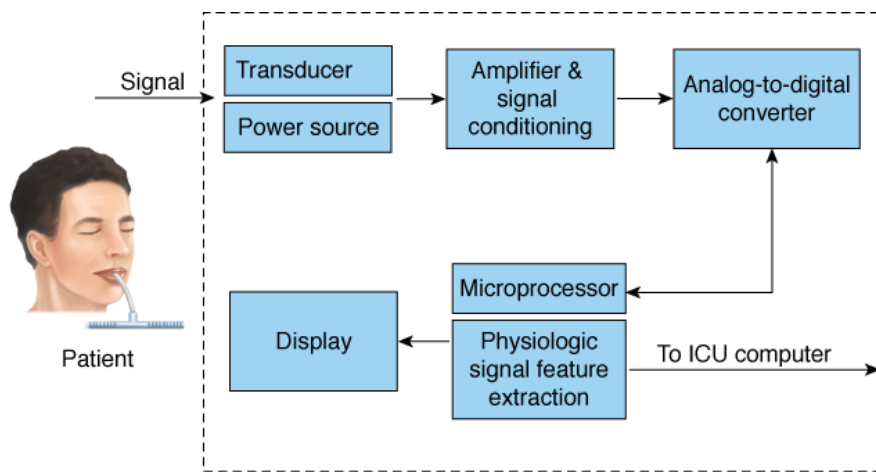
Airway pressure (Paw), esophageal pressure (Pes), and gastric pressure (Pga) in a patient with chronic obstructive pulmonary disease being ventilated with assist-control ventilation. The airway was occluded during expiration in order to quantify intrinsic PEEP (PEEPi). The patient made an expiratory effort during the occlusion (arrow), and thus the recorded pressure overestimated the true PEEPi.

Patient cooperation is also needed for reliable measurement of maximal inspiratory pressure, a measure of respiratory muscle strength.³⁷ In patients with low respiratory drive, coaching patients to make vigorous inspiratory efforts led to a 28% increase in pressure.³⁸

Fidelity of Recordings

Figure 48-15 shows the typical configuration of a data collection system used for bedside monitors in the ICU.³⁹ The physiologic quantities being measured are referred to as *signals*. A signal (blood pressure, respiratory rate) is converted from a mechanical format to an electrical format using a transducer. The signal is then amplified and conditioned. Next, it is converted from an analog signal to a digital signal with an analog-to-digital converter. Once digitized, the signal is transmitted through a microprocessor. The signal then appears on the monitor screen as a waveform or digital display.³⁹

Figure 48-15



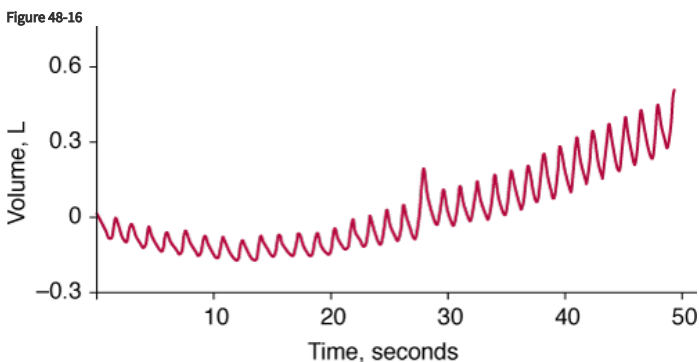
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Schematic of a bedside monitor data-acquisition system. (Modified, with permission, from Gardner.³⁹)

Numbers appearing on the screen of a monitor do not provide all of the information needed to care for a critically ill patient. Studies have revealed that data from bedside monitors account for only 13% to 22% of the data needed to make medical decisions in the care of a critically ill patient.⁴⁰

Of the other data used in clinical decision making, 21% to 22% is derived from observations of a patient by physicians, nurses, and other staff, 38% to 41% from laboratory data, and 13% to 23% from information on medication usage. Caring for a critically ill patient requires the incorporation of all of these data into clinical judgments. Ignoring data that are not important or simply noise is one of the major tasks for a caregiver in the ICU.

Acquiring high-fidelity (minimal amount of noise) data involves the following three factors. First, the transducer and data collection equipment needs to be well designed. For example, to detect the development of dynamic hyperinflation by means of monitoring of changes in end-expiratory lung volume, one cannot employ a pneumotachograph that exhibits baseline drift (Fig. 48-16). To help in the design of reliable equipment, several standards have been established by organizations such as the Association for the Advancement of Medical Instrumentation (AAMI) and the American National Standards Institute (ANSI).

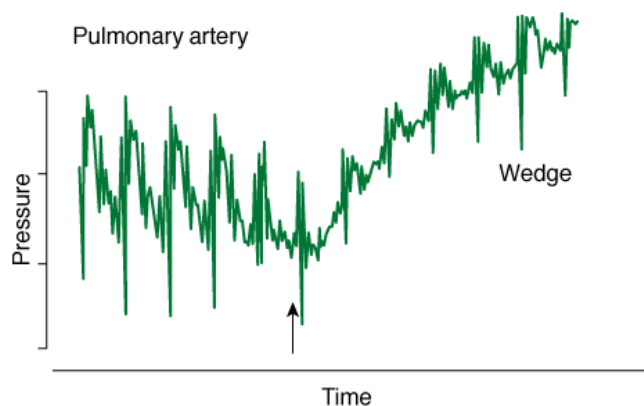


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Breath-by-breath measurements of volume obtained with a pneumotachograph in a spontaneously breathing patient. The drift in the baseline of the signal means that the pneumotachograph does not provide a reliable measure of breath-by-breath change in end-expiratory lung volume, which is desired for detecting the development of dynamic hyperinflation.

Second, the clinical staff must properly carry out certain procedures. A perfect catheter, transducer, and flush system can be set up to measure PA wedge pressure. If the catheter, however, is not positioned in the proper location in the PA, overwedging or incomplete wedging can occur, leading to erroneous readings (Fig. 48-17).

Figure 48-17



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Pulmonary artery pressure tracing demonstrating overwedging. The *arrow* indicates the point of balloon inflation.

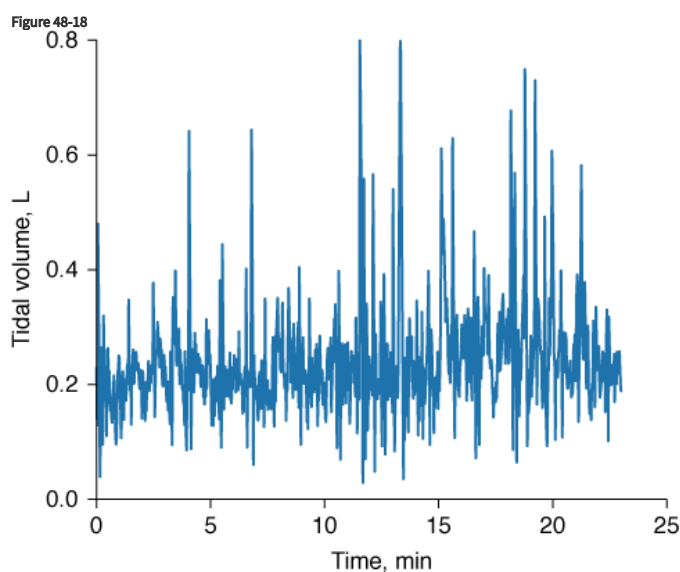
Third, rigorous and appropriate processing of the signals generated by a well-designed transducer can eliminate noise in physiologic signals. When signal extraction technology was employed in a pulse oximeter, a 22-fold reduction in noise secondary to gross motion and tremors was observed.⁴¹

Other Barriers to Accurate Data Gathering

Factors other than accuracy of the measurement can lead to faulty interpretation by the clinician.

Physiologic Variation

Even when a patient is in a steady state, variables monitored in the ICU may not be consistent. To know if a new recorded value of a variable is truly different from an earlier recorded value, it is necessary to know the normal physiologic variation of the variable (Fig. 48-18).¹⁹ Breathing pattern varies considerably. Breath-to-breath variability in tidal volume, as reflected by coefficient of variation, was $33.0 \pm 14.9\%$ (SD) in forty-seven young healthy subjects (ages 21 to 50 years) and significantly higher in eighteen healthy older subjects (ages 60 to 81 years), $44.0 \pm 14.7\%$.⁴² The substantial variability in breathing pattern between subjects, which is further magnified by the considerable breath-to-breath variability within subjects, poses a problem in selecting settings during mechanical ventilation. In a patient being ventilated with the volume-cycled form of assist-control ventilation, if tidal volume is set at the average value during unassisted breathing (approximately 5 to 7 mL/kg), the wide breath-to-breath variation in ventilatory demand means that around half of the delivered tidal volumes will be lower than the patient's demand and may lead to dyspnea.⁴³



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Breath-by-breath measurements of tidal volume in a patient during a weaning trial. Average tidal volume was 0.23 L. Note the considerable breath-by-breath variability (coefficient of variation was 43%).

Arterial pressure varies throughout the respiratory cycle. Minute-to-minute variation in systolic pressure (quantified as standard deviation) is approximately 4 mm Hg, and diastolic pressure varies 2 to 3 mm Hg. In healthy subjects, arterial pressure differs by as much as 20 mm Hg between the two arms.⁴⁴

Consequently, a difference of 20 mm Hg between the two arms does not necessarily indicate an underlying disorder such as aortic dissection.

Inherent Limitations of Monitors

No machine, including a monitor, is 100% reliable. Despite its undisputed popularity, pulse oximetry fails to provide valid measurements of Sa_{O_2} in a variety of settings.⁴⁵ The incidence of overall failure (defined as at least one continuous gap in data that exceeded 10 minutes) in anesthetized patients in an operating room was 9%.⁴⁶

Complexity of Measurements

Many users of monitors have a limited understanding of how a machine functions and the implications of its measurements. Iberti et al⁴⁷ found that 47% of physicians (77% of whom were in training) were unable to interpret a straightforward recording of a PA wedge pressure. On average, physicians incorrectly answered ten of thirty-one questions dealing with insertion and complications, cardiac physiology, and interpretation and application of data. More respondents had difficulty with questions regarding data interpretation and patient treatment than with questions related to catheter insertion or cardiac physiology.

Modern ventilators provide a continuous online display of physiologic signals (such as flow, volume, and airway pressure) and numeric computations (resistance, compliance, and work of breathing). The complexity of the physiologic principles that underpin respiratory mechanics, however, has impeded the use of the measurements in clinical practice. The poor understanding of lung mechanics by medicine residents was highlighted in the recent survey.⁴⁸ Some 35% of residents were unable to select a ventilator setting that would decrease PEEPi in a patient with chronic obstructive pulmonary disease (COPD) who had become hypotensive immediately after intubation (Table 48-1).

Table 48-1: Medical Resident's Knowledge of Mechanical Ventilation

Test items	% Incorrect
Appropriate use of PEEP in hypoxemia	44
Identifying patient's readiness for a weaning trial	38
Managing auto-PEEP correctly	35
Identifying candidate for noninvasive mechanical ventilation	27
Diagnosing tension pneumothorax	14

Abbreviation: PEEP, positive end-expiratory pressure.

The mountains of data generated by monitors make it extremely difficult to differentiate relevant from irrelevant information. More than 230 new data points may be available for review in a single patient on morning ICU rounds.⁴⁹ Psychologists have shown that the human mind is capable of assimilating no more than seven variables simultaneously.⁵⁰ When the same variable is measured by more than one machine (heart rate recorded by an ECG monitor and a pulse oximeter), the problem is compounded.

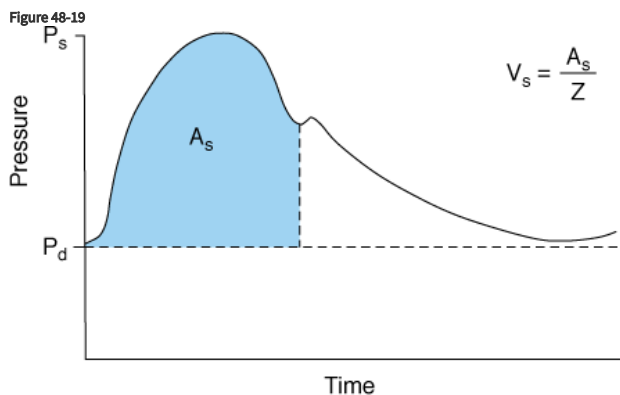
Monitoring the Right Physiologic Phenomenon

A major purpose of monitoring is to measure key variables that enhance understanding of underlying pathophysiology. Because of technical limitations, however, we tend to monitor that which we can rather than seeking the information we need.^{2,51}

The primary goal of mechanical ventilation is to rest the respiratory muscles. Thus, activity of the respiratory muscles is the single variable most needed when monitoring a ventilated patient. Ventilator settings, however, are usually adjusted on the basis of arterial blood gases. Yet, arterial blood gases provide zero information about whether the respiratory muscles are being adequately rested. Consider synchronized intermittent mandatory ventilation, which has always been adjusted according to arterial blood gases. The failure of arterial blood gases to provide insight into work of breathing probably accounts for the gap of 20 years between the introduction of this mode and recognition of its harmful effects.⁵²⁻⁵⁴

Investigators recently have argued that calculations derived from the arterial pressure waveform provide a reliable continuous estimate of cardiac output (Fig. 48-19).⁵⁵⁻⁵⁷ In patients with sepsis, Monnet et al⁵⁸ compared the accuracy of calculations based on the arterial pulse contour against a new transpulmonary thermodilution technique (the reference measurement) in detecting short-term changes in cardiac index (<1 hour). The contour technique was reliable in detecting a 15% increase in cardiac index induced by volume expansion (area under the receiver operating characteristic [ROC] curve, 0.87) or by

norepinephrine (area under ROC curve, 0.92). Although promising, the ability of the arterial pressure contour to reliably estimate cardiac output needs further investigation in a more heterogeneous group of ICU patients (not just sepsis) and over a longer period of observation (>1 hour).



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Schematic of the arterial pressure waveform versus time. P_s represents the systolic pressure, P_d the diastolic pressure, and A_s area under the systolic portion of the pressure-time curve. The heart's stroke volume V_s is related to A_s and the vascular impedance (Z). (Used, with permission, from Mahutte.⁵⁷)

Thinking in terms of ventricular preload, knowledge of end-diastolic volume is pivotal for logical assessment of hemodynamic performance in a critically ill patient.⁵⁹ No simple method is available, however, to measure end-diastolic volume. Many clinicians use central venous pressure to assess preload despite numerous studies showing that central venous pressure is extremely unreliable in estimating end-diastolic volume in critically ill patients.^{60–65}

Different Techniques for Measuring the Same Physiologic Process

In the ICU, different techniques can be used to measure the same physiologic function. The resulting differing numbers, however, may suggest different clinical states. Arterial oxygenation can be assessed by measuring O_2 saturation with a pulse oximeter or partial pressure of oxygen (P_{O_2}) by blood-gas analysis. In a study of patients with acute respiratory distress syndrome (ARDS), change in Sp_{O_2} moved in an opposite direction to the change in Pa_{O_2} in 25% of the measurements. In some patients, a decrease in partial pressure of arterial oxygen (Pa_{O_2}) of 20 mm Hg was accompanied by a simultaneous increase in Sp_{O_2} of 20%.⁶⁶ Therapeutic decisions based on Sp_{O_2} alone differed from decisions based on Pa_{O_2} on 16% of occasions.

Clinical Applications

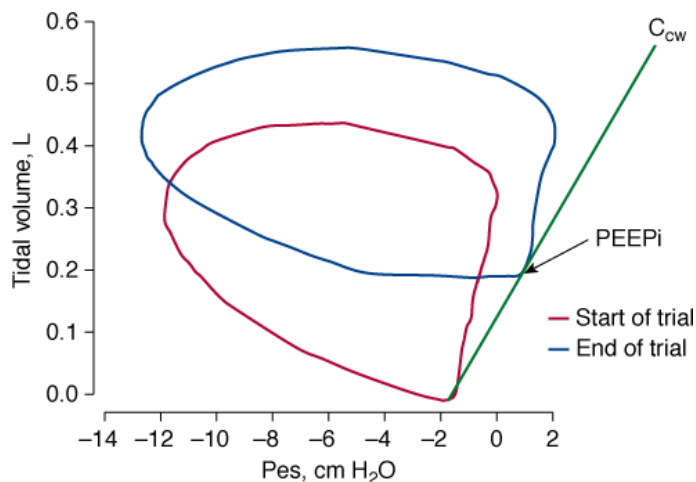
The range of options for monitoring a ventilated patient is enormous and beyond the scope of this chapter. A few examples are provided. For a comprehensive review of the clinical applications of ICU monitoring, the reader is referred to the companion book, *Principles and Practice of Intensive Care Monitoring*.⁴

Enhance Understanding of Pathophysiology

To properly interpret information—medical or nonmedical—a person must take into account the relevant context. To properly understand the results of a diagnostic test, a clinician requires a good understanding of pertinent pathophysiology.

Measuring pressure, flow, and volume during mechanical ventilation assists in the differential diagnosis of respiratory failure. The airway occlusion technique makes it possible to carefully characterize the mechanics of the lung, the chest wall, and the total respiratory system.^{67–70} During a weaning trial, esophageal pressure and flow measurements can be used to partition patient effort into its resistive, elastic, and PEEPi components.^{71,72} All three components are increased in weaning-failure patients, with PEEPi increasing the most. Therapy to decrease these abnormalities (such as bronchodilators or diuretics) may help expedite weaning outcome (Fig. 48-20).^{73,74}

Figure 48-20

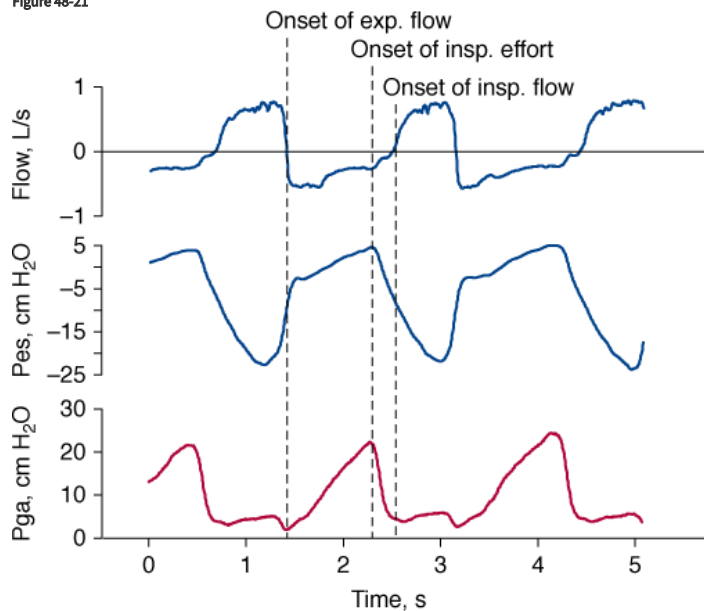


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Esophageal pressure (P_{es}) versus tidal volume curves (Campbell diagrams of work of breathing) in a patient at the start (*red*) and end (*blue*) of a trial of spontaneous breathing. The *green line* represents the static compliance of the chest wall (C_{CW}) measured during controlled mechanical ventilation. At the end of the trial, the P_{es} -volume curve is displaced upward, indicating an increase in end-expiratory lung volume as reflected by the presence of intrinsic positive end-expiratory pressure ($PEEP_i$).

Recording expiratory muscle activity during spontaneous breathing helps differentiate $PEEP_i$ caused by dynamic hyperinflation from that caused by expiratory muscle activity (Fig. 48-21).⁷⁵⁻⁷⁸ If $PEEP_i$ is mostly caused by dynamic hyperinflation, addition of external PEEP is likely to decrease work of inspiration.^{71,79} If, however, an increase in $PEEP_i$ is caused by increased activity of the expiratory muscles,⁸⁰ addition of external PEEP will impose an additional elastic load. More importantly, it will also increase the operating lung volume.

Figure 48-21



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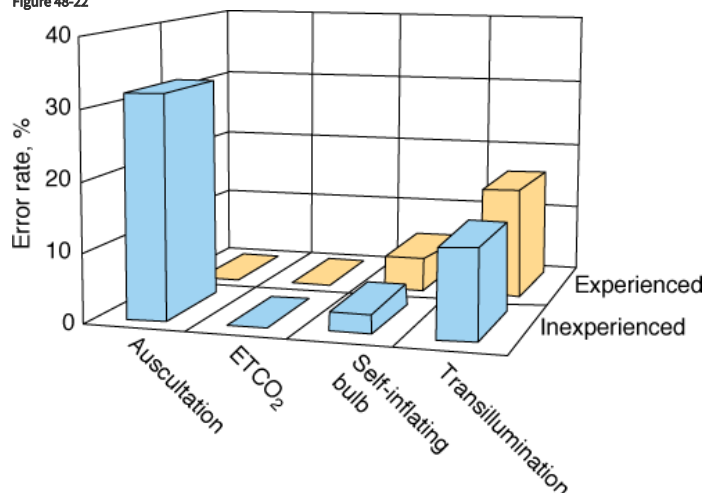
Flow, esophageal pressure (P_{es}), and gastric (P_{ga}) pressure in a patient during a trial of spontaneous breathing. Dynamic intrinsic positive end-expiratory pressure ($PEEP_i$) is estimated as the decrease in P_{es} between the onset of inspiratory (*insp.*) effort (*second vertical line*) and the onset of inspiratory flow (*third vertical line*). Estimation of the expiratory muscle contribution to swings in P_{es} and dynamic $PEEP_i$ is obtained by measuring the increase in P_{ga} between the onset of expiratory flow (*exp.*; *first vertical line*) and the onset of inspiratory effort (*second vertical line*). (Used, with permission, from Parthasarathy et al.⁷⁸)

Aid with Diagnosis

Capnometry helps in detecting esophageal intubation.⁸¹⁻⁸³ The use of end-tidal CO_2 monitoring is based on the premise that CO_2 recorded at the end of a tidal breath reflects arterial partial pressure of carbon dioxide (P_{CO_2}). When the trachea of a patient with an intact pulmonary circulation is intubated, end-tidal CO_2 is recorded. When, however, the endotracheal tube is erroneously placed in the esophagus, end-tidal CO_2 is zero. Monitoring of end-tidal CO_2 has been

compared with three other methods (auscultation, a negative pressure test using a self-inflating bulb, and transillumination) for verifying tracheal tube placement.⁸⁴ (The second approach, the negative-pressure test, uses a self-inflating bulb: If the endotracheal tube is in the esophagus, suction on the bulb causes the pliable esophageal wall to collapse and no air can be aspirated; if the tube is in the trachea, the semirigid walls prevent tube occlusion when suction is applied and air can be aspirated.) Monitoring of end-tidal CO₂ was found to be the most reliable (Fig. 48-22).

Figure 48-22

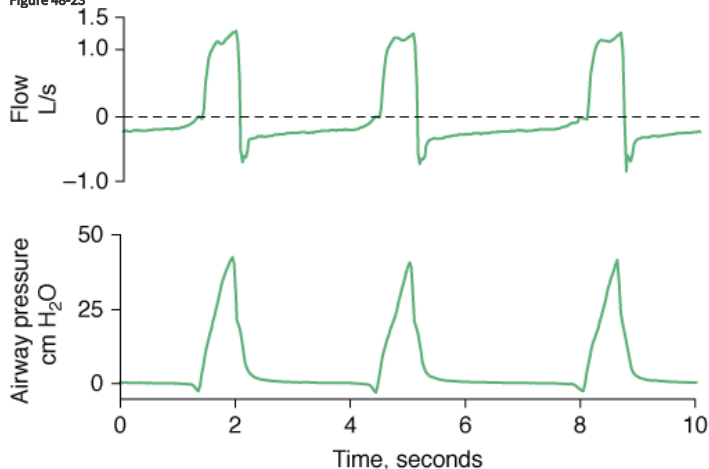


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Error rates of experienced and inexperienced examiners in verifying the placement of a tracheal tube using four methods: *auscultation*, capnographic determination of end-tidal CO₂ (ETCO₂), negative-pressure test using a *self-inflating bulb*, and *transillumination* using a lighted stylet. Capnography was the most reliable method regardless of the examiner's experience. (Used, with permission, from Knapp et al.⁸⁴)

Persistence of expiratory flow throughout expiration, without return to zero, suggests PEEPi (Fig. 48-23); this suspicion can be verified by occluding the expiratory port of the ventilator immediately before the onset of the next breath.⁸⁵ Occlusion causes pressure in the lungs and ventilator circuit to equilibrate. The pressure displayed on the ventilator manometer is PEEPi.⁸⁶ The presence of PEEPi can help explain an unexpected drop in cardiac output or electromechanical dissociation (during cardiopulmonary resuscitation).⁸⁷ Steps to overcome PEEPi can decrease patient work of breathing and improve triggering of the ventilator.^{86,88,89}

Figure 48-23



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Flow and airway pressure in a patient being ventilated using assist-control ventilation. The failure of expiratory flow to return to zero at the onset of the successive mechanical breaths indicates the presence of intrinsic positive end-expiratory pressure (PEEPi).

Guide Management

Assess the Response of Drugs

Monitoring is essential with administration of therapeutic agents that can produce rapid and dramatic changes in a patient's condition. A physician would not dream of administering a potent, rapid-acting intravenous vasoactive agent (vasodilator, vasopressor, or inotropic agent) without monitoring arterial pressure.

It would be unconscionable to perform cardiac resuscitation without ECG monitoring.

Measurement of airway resistance helps in assessing the response to bronchodilator therapy. In ventilator-dependent patients with COPD, delivery of albuterol from a metered-dose inhaler produces a decrease in airway resistance within 5 minutes; the effect is sustained for at least 60 minutes.^{90,91}

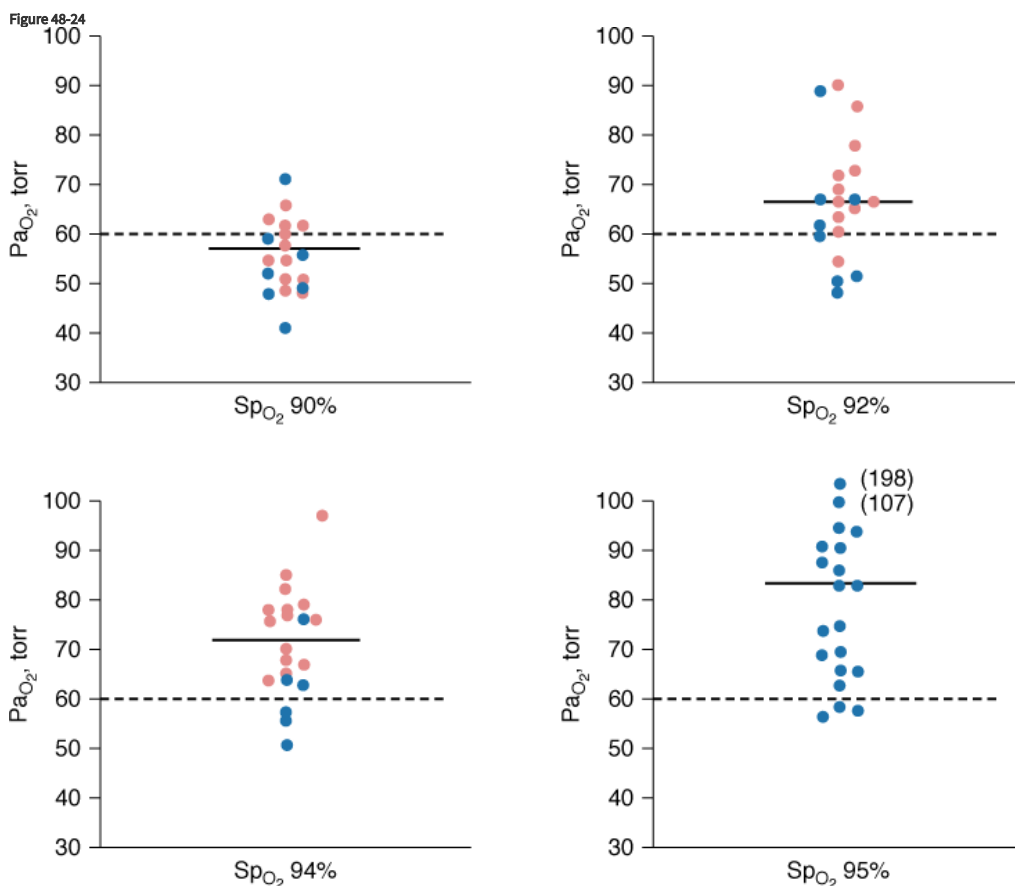
During mechanical ventilation, pulse pressure (the gradient between systolic pressure and diastolic pressure) is maximal (PPmax) at end-inspiration and minimal (PPmin) during expiration. These respiratory variations in pulse pressure have been shown to reflect changes in left-ventricular stroke volume.⁹² Accordingly, it has been reasoned that fluid responsiveness may be assessed by estimating the respiratory changes in pulse pressure (referred to as ΔPP) in terms of the following formula:

$$\Delta PP = \left[\frac{PP_{\max} - PP_{\min}}{(PP_{\max} + PP_{\min})/2} \right] * 100$$

ΔPP was found to be reliable in identifying fluid responsiveness in sedated patients receiving mechanical ventilation; this variable, however, was inaccurate in spontaneously breathing patients and in patients ventilated with low tidal volume (<8 mL/kg).^{93,94}

Optimize Ventilator Setting

Titrating FiO_2 Pulse oximetry is commonly used when titrating FiO_2 in ventilated patients (Fig. 48-24). A SpO_2 of 92% is a reliable target in white patients. In black patients, however, an SpO_2 of 92% is associated with significant hypoxemia, and a higher target, 95%, is required.⁹⁵



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Values of arterial oxygen tension (Pa_{O_2}) at pulse oximetry oxygen saturation (Sp_{O_2}) readings of 90%, 92%, 94%, and 95%. Inspired oxygen concentration (FiO_2) was adjusted to achieve a particular steady-state Sp_{O_2} value in a group of fifty-four critically ill patients. The solid horizontal line represents the mean Pa_{O_2} value obtained at each Sp_{O_2} target. The orange and blue circles represent values obtained in white and black patients, respectively. In white patients, a Sp_{O_2} target of 92% resulted in a satisfactory level of oxygenation, whereas a higher Sp_{O_2} target, 95%, was required in black patients. (Modified, with permission, from Jubran and Tobin.⁹⁵)

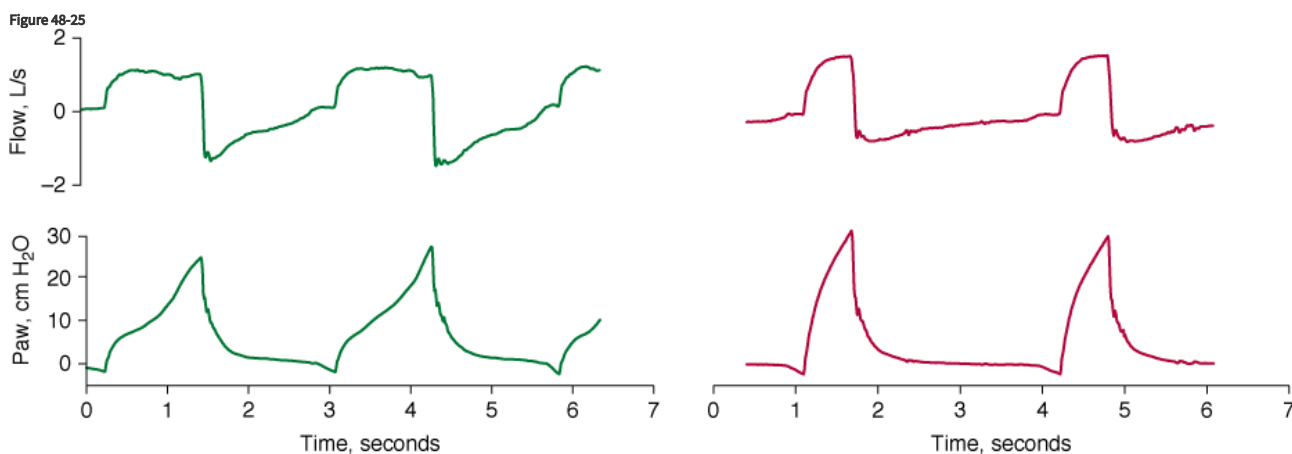
Adjusting Pressure Support. Tidal volume and respiratory rate are commonly used to set pressure-support ventilation.^{96–100} A reasonable level of inspiratory effort is an inspiratory pressure-time product less than 125 cm H₂O·sec/min. Using this target, a respiratory rate of 30 breaths/min and tidal volume of 600 mL

resulted in the fewest false classifications.¹⁰⁰

Setting Positive End-Expiratory Pressure. In a patient with acute lung injury, the right level of PEEP is that which optimizes arterial oxygenation without causing O₂ toxicity or ventilator-induced lung injury.^{101–104} Balancing the benefit of keeping the lung open (during tidal ventilation) against the risks of lung overinflation may require monitoring of the pressure-volume curve, lung morphology, and gas exchange.^{105–110}

When loss of aeration has a focal distribution (atelectatic lower lobes coexisting with aerated upper lobes), a high level of PEEP can cause overinflation of already aerated areas and only partial recruitment of atelectatic areas.¹¹¹ In such patients, some experts recommend limiting PEEP to low levels (approximately 10 cm H₂O).¹⁰¹ In patients who have diffuse loss of aeration, the risk of overinflation with PEEP is low.¹¹² Thus, a higher PEEP can be used provided it does not cause the plateau pressure to rise above the upper inflection point of the pressure-volume curve.^{101,113}

Assessing Patient Work of Breathing. Inspection of the airway pressure contour during assisted ventilation can help assess patient work of breathing. Excessive scalloping of the airway pressure tracing indicates increased work, such as that caused by insufficient inspiratory flow (Fig. 48-25). During assist-control ventilation, an increase in flow can decrease work of breathing by as much as 60% in patients with acute respiratory failure. Higher flow rates can also decrease inspiratory effort in stable patients with COPD.^{114,115}

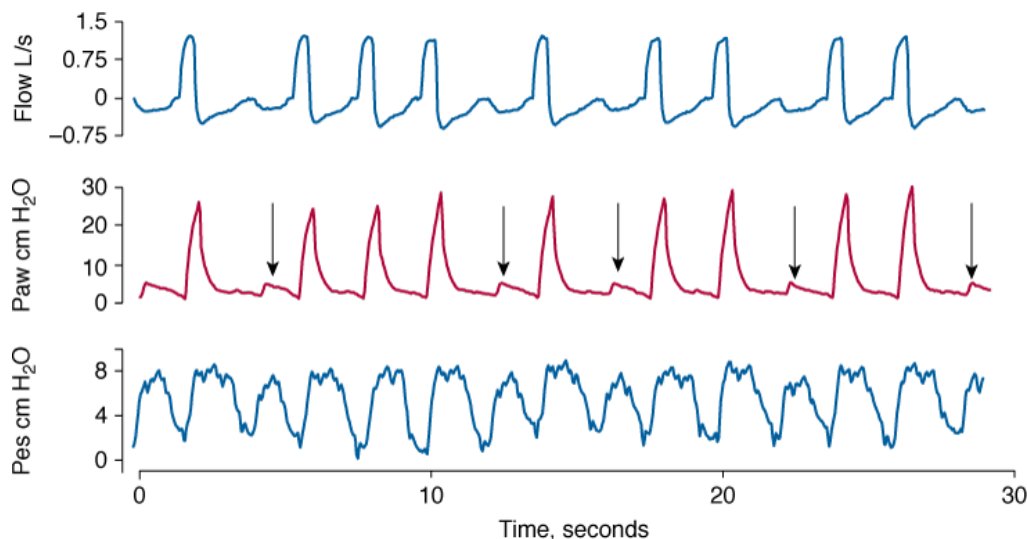


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Flow and airway pressure (*Paw*) during assist-control ventilation at a flow of 30 L/min (*left panel*) and a flow of 60 L/min (*right panel*). On the *left panel*, the pronounced negative phase together with excessive scalloping of the *Paw* waveform indicates that the patient is making a strenuous effort to breathe as a result of the inadequate flow setting. On the *right panel*, the small negative phase and smooth rise and convex appearance of the *Paw* waveform indicate that the patient is making a slight inspiratory effort to breathe.

Continuous displays of pressure and flow have led to increased awareness that patient effort is frequently insufficient to trigger the ventilator.^{75,88} During pressure support or assist-control ventilation, up to a third of patient efforts may fail to trigger the machine (Fig. 48-26). Such nontriggering has been shown to result from premature inspiratory efforts that are insufficient to overcome the elastic recoil associated with dynamic hyperinflation.⁸⁸ To trigger the ventilator, patient effort has to first generate a negative intrathoracic pressure to counterbalance the elastic recoil and then overcome the set sensitivity. The full consequences of wasted inspiratory efforts are not known. They certainly place an unnecessary burden on patients whose inspiratory muscles are already under stress. Such added stress can interfere with subsequent weaning.^{116–118}

Figure 48-26



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Flow, airway pressure (*Paw*), and esophageal pressure (*Pes*) in a patient receiving assist-control ventilation. The *arrows* indicate inspiratory attempts that failed to trigger the opening of the ventilator valve.

Avoid Complications

The heterogeneous lung involvement in ARDS puts some regions at risk of developing alveolar overdistension when a ventilator breath is delivered.^{119–121} Plateau pressure is monitored as a surrogate for end-inspiratory alveolar pressure, and its monitoring may help to minimize lung injury.¹²² Studies show that patients with ARDS who have plateau pressures greater than 32 cm H₂O have higher mortality.^{113,123,124}

Daily radiographs in ventilated patients can reveal unsuspected or significant abnormalities.^{125,126} Malpositioning of devices (endotracheal tubes, nasogastric tubes, and central venous pressure lines) and pneumothorax or pleural effusions are detected in 20% of patients.¹²⁷ Physicians had not anticipated these findings. The importance of early detection of abnormalities was underscored in a study by Steier et al.¹²⁸ Mortality was 7% when pneumothorax was diagnosed immediately. Mortality quadrupled when diagnosis (and treatment) was delayed.

Provide Alarms

Pulse oximetry can provide an early warning of hypoxemia. In 20,802 surgical patients, use of pulse oximetry resulted in 19-fold greater detection of hypoxemia (defined as an $Sp_{O_2} < 90\%$) as compared with patients in whom an oximeter was not used.¹²⁹ Myocardial ischemia was less common in the oximetry group than in the control group, twelve and twenty-six patients, respectively. The anesthesiologists reported that oximetry led to a change in therapy on one or more occasions in 10.5% of patients in the operating room and in 17% in the postanesthesia care unit.

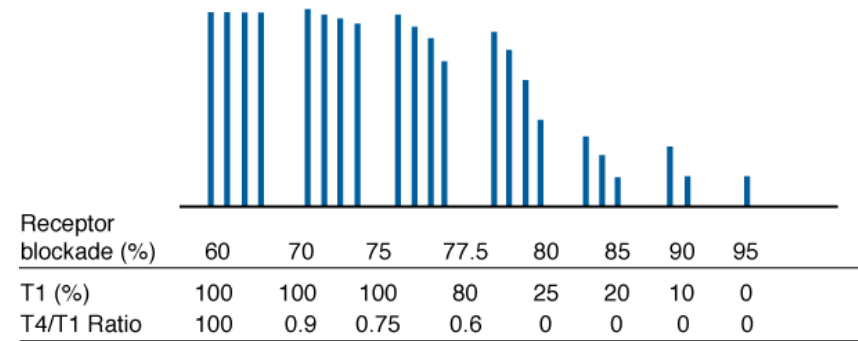
An alarm on a ventilator may sound (or flash) because of a change in ventilator performance or patient clinical status. An abrupt increase in peak airway pressure can arise with endotracheal obstruction or ventilator malfunction. A decrease in peak pressure can arise with a leak in the circuit. An increase in the baseline airway pressure can signal malfunction of the exhalation valve (or inadvertent PEEP).^{130,131}

The role of monitoring as a warning sign was highlighted in a study of anesthesia-related malpractice claims.¹³² Of 1097 cases reviewed, negative outcomes were prevented in 32% by the use of additional monitors. The authors estimated that use of a pulse oximeter and capnometer would have prevented 93% of the mishaps.¹³²

Assessment of Trends

Monitoring of physiologic variables over time helps in assessing a therapeutic response.¹³³ When vasopressor therapy is being titrated, continuous monitoring of arterial pressure is essential. Monitoring the motor response to peripheral nerve stimulation helps avoid the accumulation of neuromuscular blocking agents (with risk of prolonged paralysis).^{134,135} Train-of-four stimulation uses a series of four supramaximal stimuli delivered at a frequency of two stimuli per second (Fig. 48-27). The dose of a neuromuscular blocking agent is usually titrated to obtain no more than two visible muscle twitches in response to each train-of-four. If the patient fails to demonstrate any response with the train-of-four stimulation, the blocking agent should be discontinued temporarily or the dosage reduced.^{135–137}

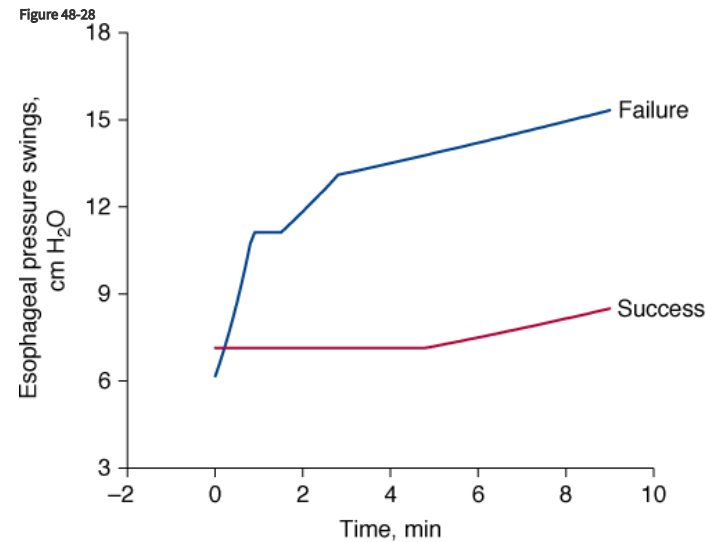
Figure 48-27



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Train-of-four stimulation consists of a series (train) of four electrical impulses delivered at a frequency of two stimuli per second. When neuromuscular blocking agents are not being used, the response to train-of-four stimulation is four strong twitches. As the receptors begin to load up with the neuromuscular blocking agent, four twitches are still present, although the response is weaker and a decrease is evident between the first and the fourth twitch. Comparison of muscular movement on the first (*T1*) and fourth twitch (*T4*) enables the calculation of a *T4/T1* ratio; this ratio can detect an earlier stage of receptor blockade than twitch loss. (Used, with permission, from Strange.¹³⁶)

Changes in esophageal pressure over the first 9 minutes of a trial, quantified as a trend index, revealed sensitivity 0.91, specificity 0.89, positive predictive value 0.83, and negative predictive value 0.94 (Fig. 48-28).⁷⁴ The area under a ROC curve for the trend index (0.94) was greater than for first-minute measurement of esophageal pressure (0.44, *p* < 0.05) and tended to be greater than that for frequency-to-tidal volume ratio (*f/V_T*).



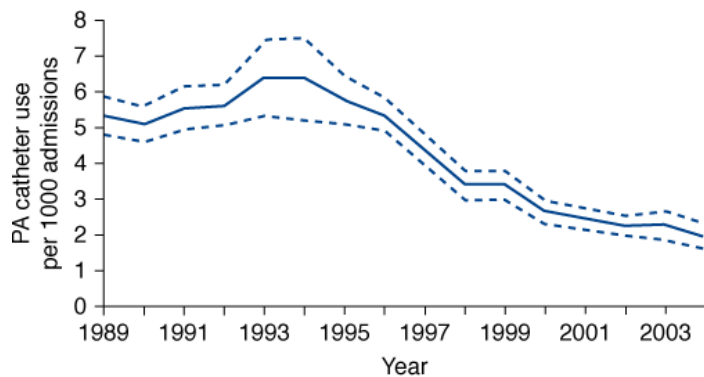
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Change in esophageal pressure (*Pes*) swings, quantified using a nonlinear model, namely the multivariate adaptive regression spline (MARS) analysis, during the first nine minutes of weaning trials in weaning-failure patients (*blue line*) and weaning-success patients (*red line*). The mathematical technique was used to capture all of the data points in the two groups over the time span. The greatest rate of increase in *Pes* swings in the failure patients occurred during the first minute. The greatest difference in *Pes* swings between the failure (14.5 cm H₂O, 95% confidence intervals 18.9 to 11.2) and success groups (7.9 cm H₂O, 95% confidence intervals 11.8 to 0.8) was at the transition between the ninth and tenth minute. (Used, with permission, from Jubran et al.⁷⁴)

Effect of Monitoring on Patient Outcome

Because of expectation of improvement in patient outcome, monitoring is subjected to closer scrutiny than most diagnostic tests.⁵¹ Although complications derived from use of a monitor are relatively easy to evaluate, few monitoring devices have been subjected to studies of their effect on mortality or other clinical outcomes. Those few studies have not revealed clearcut benefits. Three recently conducted randomized trials of the PA catheter (in 2714 patients and 120 ICUs) did not reveal improved outcome.^{138–140} The negative results of these studies have led to a substantial decline in the use of the PA catheter (Fig. 48-29).¹⁴¹

Figure 48-29



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Pulmonary artery (PA) catheters placed per 1000 patients (*solid line*) and 95% confidence intervals around the annual rate of catheter utilization (*dashed lines*) in nine American states as part of a nationwide inpatient sample. Between 1993 and 2004, use of pulmonary artery catheterization decreased by 65%, from 5.66 to 1.99 per 1000 medical admissions. (Modified, with permission, from Wiener et al.¹⁴¹)

In 20,802 surgical patients, Moller et al¹²⁹ found that pulse oximetry had no demonstrable benefit on the rate of postoperative complications. Accordingly, it is reasonable to ask whether monitoring has any role in the management of a critically ill patient. At first glance, the studies say no. Before reaching that conclusion, however, we need to scrutinize the data.

A randomized controlled trial is generally adopted as the most definitive method for evaluating the benefit of some intervention. In a typical trial, the effect of an intervention on some outcome is assessed in a well-defined patient population. Such trial designs are mostly used to assess the risks and benefits of a therapeutic agent. Using a randomized trial to assess the benefits of a monitoring device can be problematic because of poorly defined patient populations, complexity of the therapeutic intervention (because monitoring is also linked to treatment), and choice of the primary outcome.

Failure to Precisely Define Disease State

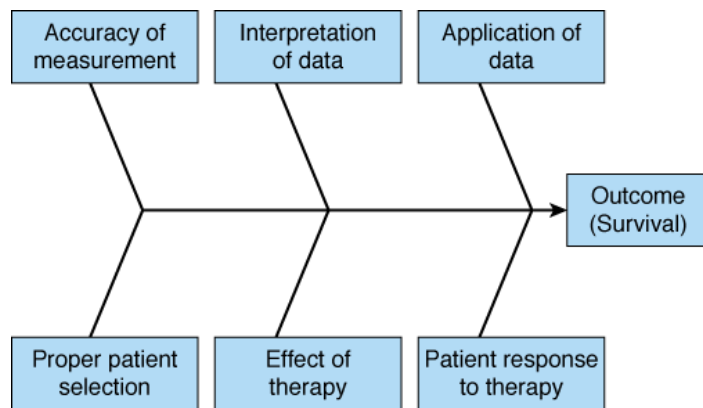
The first step in evaluating the effect of an intervention is to ensure a well-defined study population. Definition and categorization of disease states—the discipline of nosology—is especially difficult in critically ill patients.¹⁴² (The reader is referred to a detailed discussion of nosology in [Chapter 4](#).) The most precise way to define a disease is in etiologic terms (e.g., Legionnaire disease). Most randomized trials of monitoring techniques have been conducted in groups of patients identified in terms of a syndrome—that is, a condition defined on the basis of a constellation of symptoms and signs. A typical definition of sepsis is the following: temperature greater than 38°C (100.4°F) or less than 36°C (96.8°F), respiratory rate greater than 20 breaths/min or P_{CO_2} less than 32 mm Hg, and signs of hypoperfusion, among others.¹³⁹ (Note that the respiratory rate commonly listed as a criterion for sepsis falls in the middle of the normal range.) When a disease is defined in an imprecise manner, very different types of patients may be assigned to the two arms of a trial. This factor could be a major contributor to negative outcome of a study of a monitoring technique.

Complexity of Evaluating the Intervention

Evaluating the benefit and risk of any intervention in the ICU is challenging. The huge number of procedures and treatments typically used in any single ICU constitute confounding variables. These covariates contribute to considerable experimental noise in a trial, masking benefits that are truly present.

To illustrate these considerations, we discuss a hypothetical study designed to determine whether PA catheter monitoring improves survival in septic shock. In designing such a trial, the covariables shown in [Figure 48-30](#) must be controlled for or taken into account.

Figure 48-30



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Factors that can influence the effect of a monitoring device on patient outcome, such as survival. See text for details.

Patient Selection: Distinction between True Indications versus Study Inclusion Criteria

The choice of the most appropriate monitoring device is dictated by a patient's condition. To decide whether a monitor is indicated in a particular patient, the physician must balance potential benefits and risks. When wondering whether to insert a PA catheter in a patient with septic shock who is unresponsive to fluid and vasopressor therapy, a physician should consider three factors. One, will end-organ dysfunction proceed and become irreversible if therapy is not changed? Two, will large volumes of fluids, administered empirically but on an incorrect assumption that intravascular volume is depleted, predispose to hypoxemia should acute lung injury be present? Three, if a PA catheter is inserted, will the physician withhold fluids if the wedge pressure is normal?

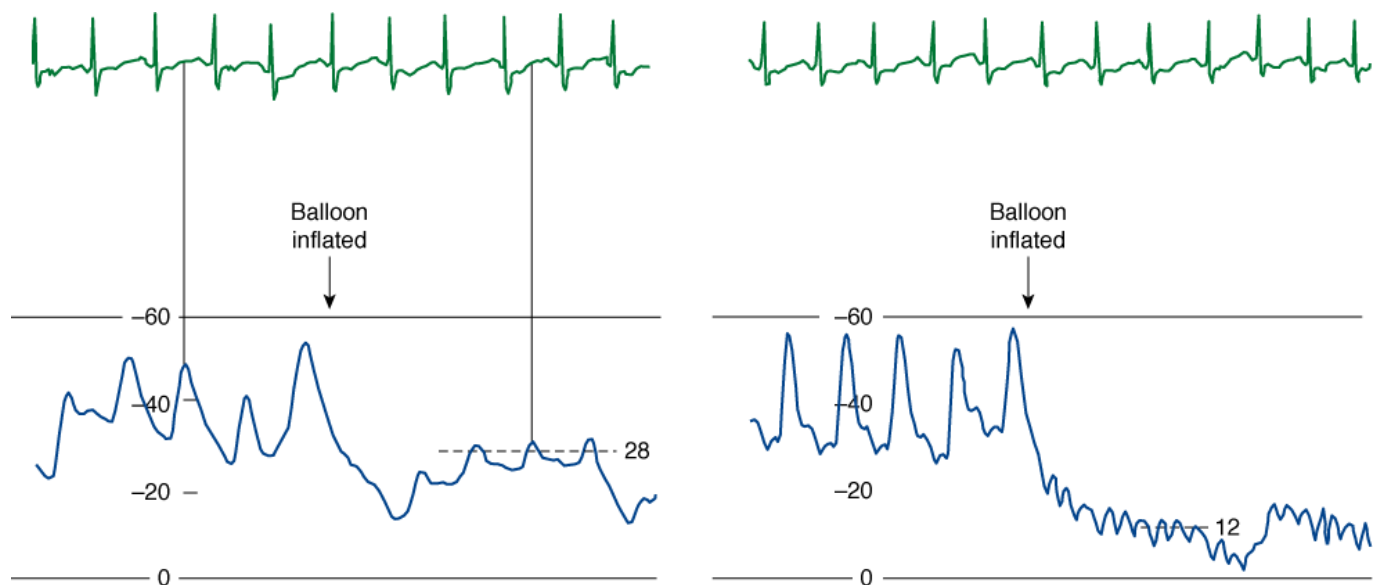
The uncertain probability of organ dysfunction and its time course, the risks of monitoring, and the variable efficacy of therapy highlights the complexity of clinical decision making, even when the clinician is highly skilled and experienced. The above three questions, and other factors that impinge on the decision of whether a PA catheter is indicated in a particular patient, are omitted from the design of a randomized clinical trial. Instead, patients are selected on the basis of carefully defined eligibility criteria. The study is thus focused on a narrowly defined spectrum of patients who poorly reflect the more heterogeneous range of disease seen in everyday practice. As a result, the patients receiving a PA catheter in a randomized trial may bear little relationship to what an experienced and cautious physician does in everyday practice. A skilled physician is unlikely to insert a PA catheter in many of the patients assigned to the intervention arm of a trial, believing that the resulting information would have little likelihood of influencing patient management. If patients are entered into a trial and are randomized to the PA catheter arm, although a judicious physician would not insert a catheter in the same patients, the consequence will be to dilute any likely benefit from the catheter but at the same time increase the risk of complications.

In a study of a mixed group of medical and surgical patients, Connors et al¹⁴³ found that a PA catheter was more likely to be used in patients who were sicker and more likely to die. Richard et al¹³⁹ studied the benefit of a PA catheter in patients with shock and ARDS. Patients were eligible for monitoring with a PA catheter if their systolic blood pressure was equal to or greater than 95 mm Hg (on 3 mcg/kg per minute of dopamine), heart rate equal to or greater than 92 beats/min, PaCO_2 equal to or less than 32 mm Hg while breathing spontaneously, and an elevated lactate level. We suspect most physicians would not insert a

PA catheter in all (or even most) patients who meet these four criteria. In the study of Sandham et al¹³⁸ in high-risk surgical patients, 87% of patients who had a PA catheter inserted were not at high risk for death (as identified by New York Heart Association or American Society of Anesthesiologists classifications). In the study by Harvey et al¹⁴⁰ of 1041 medical and surgical ICU patients, patients were randomized to the PA catheter arm if the primary physician considered the patient as someone who "should" be managed with a PA catheter; precise criteria of what constitutes "should" were not given. Again, this study did not demonstrate a benefit from a PA catheter. But if a PA catheter is inserted in a patient in whom there is no clear indication for its use, would you expect the catheter to improve outcome? We expect all readers to answer no. Thus, the outcomes in the randomized trials by Richard et al,¹³⁹ Sandham et al,¹³⁸ and Harvey et al¹⁴⁰ were all preordained as negative before the first patient was enrolled.

Accuracy of the Monitors

Studies evaluating the impact of monitoring devices on outcome are predicated on the assumption that the clinical staff are employing the monitor in a knowledgeable manner and obtaining reliable readings. Reliability of PA catheter readings is highly dependent on catheter placement, calibration of the transducers, and interpretation of the data. Of 282 measurements of PA wedge pressure, Morris et al¹⁴⁴ found that 103 (36%) of the readings were inaccurate. Of 103 erroneous readings, eighty-nine were associated with technical problems, such as damped tracings, poor dynamic response, overinflation of the balloon, or incomplete wedging (Fig. 48-31). Technical problems with use of the PA catheter, or steps taken to correct them, were not taken into account in the randomized trials conducted by Richard et al¹³⁹ and Sandham et al.¹³⁸ Specifically, Harvey et al¹⁴⁰ state, "We did not assess the quality of use." Even with great care, these technical problems cannot be eliminated completely. For a simple device, such as a pulse oximeter, overall instrument failure was 7.2% among the sicker patients in the study of Moller et al.¹²⁹ Such operational problems are commonly not listed in reports of clinical trials.



Source: Tobin MJ: *Principles and Practice of Mechanical Ventilation*, 3rd Edition: www.accessanesthesiology.com
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Simultaneous recordings of electrocardiogram (ECG) and pulmonary artery (PA) pressure to illustrate incomplete wedge pressure (Ppw). *Left panel.* With balloon inflation, wedge pressure falls to a value that approximates pulmonary artery diastolic pressure (Ppad). After balloon inflation, a single positive wave coincides with the T wave on the ECG—a pattern inconsistent with a left-atrial waveform. *Right panel.* Waveforms after the catheter has been retracted and the balloon reinflated. A large Ppad–Ppw gradient is evident and the tracing is consistent with a left-atrial waveform. Scale in mm Hg. (Used, with permission, from Leatherman and Marini.²⁰)

Interpretation of the Data

Several studies reveal that many clinicians have limited skills in the accurate interpretation of data generated by a PA catheter.^{47,145–147} In their survey, Iberti et al⁴⁷ found that almost half of physicians incorrectly interpreted a straightforward recording of wedge pressure. Another study revealed that physicians arrived at a wide range of wedge pressures for the same tracing.¹⁴⁸ If the wedge pressure is incorrectly and repeatedly read as low in a patient with ARDS, then continued administration of fluids is likely to result in an unfavorable outcome.

Application of Data in Clinical Decision Making

Even if data are accurate and properly interpreted, benefit will occur only if the physician applies the information logically in the management of patients. Some monitoring decisions can be made by simple rules of thumb (heuristics); these are easy to incorporate into an algorithm.^{149,150} Simple rules take the *if-then* form. If serum glucose in a patient with insulin-dependent diabetes is high, then **insulin** should be administered. This strategy is straightforward because the probability of hyperglycemia is high, the consequences of untreated hyperglycemia are great, the risk of monitoring (drawing a blood sample) is low, the benefit of treatment is high, and the risks when **insulin** is properly administered are small.

The monitoring dilemmas that arise in most critically ill patients are complex and cannot be captured in an algorithm. Intensivists frequently have to make rapid decisions under conditions of considerable uncertainty, involving substantial risks and unpredictable benefit. To combine, integrate, and interpret diagnostic data in their decision-making process, intensivists employ a number of strategies of clinical reasoning.

Probabilistic Reasoning

Diagnostic uncertainty arises because data are insufficient or because the relationship between the data and the disease process is poorly understood.¹⁵⁰ To cope with uncertainty, physicians may use probabilities if available.¹⁵¹ Probabilistic reasoning is based on the known sensitivity and specificity of a diagnostic (or monitoring) test. A physician forms an initial gestalt as to the likelihood of some condition, which is the *pretest probability*. Then, based on the known sensitivity and specificity of the test, the physician forms a new estimate of likelihood, the *posttest probability*. Take a physician who has no inkling as to whether a patient can (or cannot) tolerate a weaning trial. The pretest probability will be 50%. The physician then measures f/V_T and obtains a reading of 80.

This reading has a likelihood ratio of 7.5 (that the patient will tolerate a weaning trial). Based on the nomogram of Fagan,¹⁵² a likelihood ratio of 7.5 is associated with a posttest probability of 95% (if pretest probability is 50%). This means that when physicians are in complete doubt (50-50) as to whether a patient can be weaned, obtaining an f/V_T value of less than 80 will substantially decrease their doubt.

Unfortunately, the published data necessary for precise application of probabilistic reasoning are available for few diagnostic (or monitoring) techniques in the ICU. Because physicians cannot communicate in precise quantitative terms, they use expressions such as “most likely,” “unlikely,” and “rare.” Consider a

patient in septic shock who remains hypotensive despite fluid resuscitation and a moderate dose of a vasoactive agent (dopamine > 10 mcg/kg/min). A common cause of hypotension in such a patient is left-ventricular-dysfunction. To determine whether left-ventricular-dysfunction is present or absent, the physician measures the patient's cardiac index. Before measuring cardiac index, the physician forms a gestalt of the patient's likelihood of left-ventricular dysfunction (based on previous experience of patients with similar clinical characteristics). After inserting a PA catheter, measuring cardiac index, and based on the reported sensitivity and specificity of cardiac index, the clinician revises his or her assessment of the patient's likelihood of left-ventricular dysfunction. This scenario is idealistic, not realistic, because sensitivity and specificity of cardiac index for detecting left-ventricular-dysfunction in patients with septic shock have not been studied.^{153,154}

Another problem arises with use of monitored data. Although data are generally recorded as continuous variables, clinicians typically view data in dichotomous terms. What is the breakpoint in cardiac index below which a patient is classified as having left-ventricular-dysfunction? Reports on cardiac index as a diagnostic criterion of left-ventricular-dysfunction usually present a range of possible values, such as 1.6 to 2.3 L/min/m², rather than a discrete value.¹⁵⁵ The threshold value of cardiac index that is best in distinguishing between the presence and absence of left-ventricular-dysfunction has never been determined. Is it an index of 2.2 L/min/m²? Is it 2 L/min/m²? Without a defined threshold value, one cannot attempt to measure the sensitivity and specificity of a diagnostic test.

A further layer of complexity arises with the choice of reference standard. All tests are inaccurate to varying degrees. Thus, the degree of accuracy of a diagnostic (or monitoring) test is measured against a reference test. Ideally, the reference standard should be measurable in concrete terms: histology or precise level of narrowing on angiography. For left-ventricular-dysfunction, however, the reference standard is based on clinical criteria. Studies suggest that a cardiac index of 2.2 L/min/m² is an important threshold in predicting the outcome of patients with acute myocardial infarction.¹⁵⁵ Will the physician use the same cardiac index threshold in deciding whether to start treatment with dobutamine in the above-described patient with sepsis who has failed dopamine therapy? Moreover, in making decisions in a particular patient, a clinician may decide to raise or lower the threshold value, taking into account aspects unique to a particular patient's condition. Because of dobutamine's chronotropic action on the heart, a physician may decide to withhold this drug in a patient with underlying coronary artery disease until the cardiac index falls to 1.8 L/min/m².

There is a deeper, more fundamental problem with probabilistic reasoning. This relates to its epistemologic grounding. (Epistemology is the branch of philosophy concerned with the study of the very foundation on which our knowledge rests.) On their own, data derived from randomized trials are unable to explain *why* a particular relationship exists. In contrast research in physiology, molecular biology, and many other fields has as its objective the elucidation of underlying mechanisms (explanations for why relationships occur). Tanenbaum¹⁵⁶ undertook an ethnographic study of the reasoning employed by physicians as they made clinical decisions. She found that the clinicians were primarily realists (or determinists): They based decisions on a mental picture of a world of real relationships, real events. Data from mechanistic research enrich this mental picture, offering explanations (meaning) on which physicians can base clinical decisions. When physicians operate as realists, they conduct themselves as if they know what is really happening inside a sick person. Data from a randomized trial simply show that some relationship was observed, but provide no epistemologic basis for its occurrence, no understanding of cause and effect. In clinical decision making, realists image a disease process; probabilists play the odds.

Physiologic Reasoning

To attain greater confidence in diagnosis, clinicians employ physiologic (or causal) reasoning. As discussed under “[Probabilistic Reasoning](#)” above, the hope is to form a more coherent picture of cause-and-effect relationships in a given patient. This type of reasoning is particularly suited to the ICU, because monitoring is geared toward detecting changes in pathophysiology. Causal reasoning involves testing and validating (or refuting) apparent cause-and-effect connections.

Consider a patient with COPD and congestive heart failure who is being ventilated. A PA catheter is inserted and reveals a wedge pressure of 28 mm Hg. Based on probabilistic considerations, the likelihood of pulmonary edema is great. Using causal (physiologic) reasoning, wedge pressure is viewed as a reliable estimate of preload (left-ventricular end-diastolic volume). In this patient, the clinician judges the elevated wedge pressure to signify hypervolemia. The true advantage of causal reasoning becomes more clear when we learn that a 60-second interruption of positive-pressure ventilation caused this patient's wedge pressure to fall to 15 mm Hg. Probabilistic reasoning would steer us toward diagnoses such as hypervolemia or cardiac dysfunction. Physiologic (or causal) reasoning leads us to suspect that the high wedge pressure (28 mm Hg) is caused by PEEP rather than hypervolemia or cardiac dysfunction.

Causal reasoning forms the bedrock for many diagnostic decisions made outside the ICU. Our understanding of the pathophysiology of critical illnesses, however, is too rudimentary to apply it when managing most ICU patients. The above example highlights the need for much more research into pathophysiologic mechanisms in critically ill patients. Data generated by randomized, controlled trials can never provide the information needed for causal reasoning—the type of reasoning that is dominant in the decisions made by physicians in their everyday practice.¹⁵⁶

Deterministic Reasoning

Deterministic or categorical reasoning is used to reduce information overload. Deterministic reasoning applies a set of rules that take the form of *if-then* statements. For example, *if* a ventilated patient with COPD develops a sudden and equivalent increase in peak and plateau pressures, *then* a nonairway problem, such as a pneumothorax, should be considered. If a patient develops acute renal failure without proteinuria and the fractional excretion of sodium is less than 1%, then the renal failure is prerenal in origin.

Such *if-then* rules look simple and attractive. The difficult part is figuring out if the antecedent condition (the *if* part of the rule) has been met. When the developers of clinical protocols incorporate such rules into an algorithm, they deliberately eliminate contingencies that can influence the application of the rule. The elimination of such ambiguities means that the rule can be more precisely and concisely expressed on paper. But the elimination of context means that the rule is artificial and cannot be applied in a consistent manner during routine clinical practice.

An algorithm is available for treating hypokalemia in an ICU patient receiving diuretics. If serum potassium is 3.5 mEq/L and the patient is asymptomatic, the algorithm dictates not to give a potassium supplement. The context of everyday ICU practice, however, is more complex. If the patient has heart disease and is taking digitalis or undergoing cardiac surgery, most clinicians would administer potassium if serum potassium is 3.5 mEq/L to prevent life-threatening arrhythmias. Likewise, if a patient has cirrhosis, physicians would treat a potassium of 3.5 mEq/L to minimize the risk of hepatic coma. The problem of hypokalemia is further complicated if there is concurrent hypomagnesemia, which promotes potassium wasting. It would be difficult if not impossible to capture all such nuances in the few steps of an algorithm.

Research studies of monitoring devices have included algorithms in an attempt to control for variations in clinical practice. These attempts did not achieve their desired goals. Gattinoni et al.¹⁵⁷ undertook a study to evaluate the benefit of directing fluid and pharmacotherapy to achieve a supranormal cardiac index (guided by a PA catheter). Survival was equivalent in the groups with supranormal and normal cardiac index. At first sight, the findings suggest that targeting therapy to a supranormal cardiac index does not improve outcome. Perhaps that is true. But such a conclusion cannot be rationally defended based on the presented data. The target of supranormal cardiac index was reached in only 46% of patients in that study arm. Canadian researchers strove for a similar goal.¹³⁸ They compared therapy based on predefined physiologic targets (guided by a PA catheter) versus therapy based on usual care (without a PA catheter). The O₂ delivery target in the experimental arm was reached in only 21% of patients at study entry and in 63% of patients after surgery. These two studies provide a vivid illustration of the problems that investigators face in achieving the physiologic objectives that are the true focus of their randomized trial. If the target is not being achieved, what exactly is being studied in these clinical experiments?

Casuistry

Physicians face many decisions that have never been the subject of a randomized, controlled trial. A physician with many years of experience will relate a current patient to previously similar cases that he or she has handled, paying close attention to how the present patient is similar to, or differs from, the previously managed cases. The physician may also draw on knowledge gained from case reports and observational studies—the only type of literature that pertains to many problems in the ICU. Few randomized controlled trials have specifically addressed fluid management of patients with ARDS. A number of observational studies have been conducted,^{158,159} and knowledge gained from such reports combined with previous experience guide decisions on fluid management in this setting.

Reasoning along these lines, primarily in terms of analogy, is referred to as *casuistry*. Casuistry literally means being concerned with concrete individual cases, as opposed to abstract generalities.^{160,161} Casuistry involves reasoning based on precedent (previous example). It is the type of reasoning used in case law, and is the main foundation of the legal system. In deciding how best to manage a particular patient, a physician also needs to take into account a patient's personal preferences.

Tacit or Implicit Knowledge

Physicians make many decisions using tacit knowledge: the type of inexpressible knowledge in *knowing how* versus *knowing what*. Tacit knowledge is knowledge that cannot be articulated. It is implicit knowledge based in actions rather than in conscious thoughts.¹⁶² For more than 80% of complex situations, cognitive psychologists have found that experts such as intensive care staff did not employ an analytical approach to decisions. Instead, they relied on tacit knowledge.¹⁶²

The term *tacit knowledge* was introduced by Michael Polanyi, who pointed out that human beings know more than they can impart.¹⁶³ Tacit knowledge is acquired without the intervention of explicit reasoning. It suggests an activity that takes place below the level of conscious reasoning, whether by habit, by being deeply ingrained, or by instinct. By definition, its content is indefinable.

Examples of tacit knowledge include the surgeon's knowledge of the right amount of tension to place on a suture, knowing when to intubate a patient in respiratory distress, and knowing when to extubate a patient after completion of a weaning trial. The skill in performing bronchoscopy is another example. The physician knows how to position the bronchoscope spatially as if it were an extension of his or her fingertips, and how to serially weave the instrument to enter the posterior segment of right lower lobe bronchus. An added dimension is knowing how much pressure to exert, and sensing resistance, when performing a transbronchial biopsy. The physician cannot communicate this knowledge in explicit terms. Instead, the knowledge is something the physician feels and shows while performing the procedure.

The failure of the possessor of tacit knowledge to identify and communicate its content is often attributed to poor power of introspection. When a physician is asked to recount the mental steps involved in solving a complex problem, the reconstruction often sounds unconvincing, even naïve. An attending physician may decide not to extubate a patient who is breathing at a rate of 28 breaths/min and who has reasonable arterial blood-gas values. The fellow may appear puzzled by this decision. To explain his (or her) decision, the attending physician may use words such as "intuition" or "gut feeling." The word *intuition* is loaded with negative connotations. It is equated with random guessing, confused superstition, mystical thought, weird unfathomable emotion, and an excuse for prejudice. Yet, intuition, not explicit logical reasoning, lies at the heart of originality in science.

People are not born with some free-floating gift of intuition. Instead, the unconscious decisions and automatic actions that reflect tacit knowledge stem from years and years of experience and reflection in a particular field. Intuition is needed to know *when* is the right time to adapt (to judge when is a plan falling apart); to know *how* to adapt (hatching a scheme on the hoof); and in *evaluating* adaptive steps (spotting new steps that may cause problems subsequently). Tacit knowledge is exemplified by an experienced clinician who knows at what juncture it is prudent to insert a PA catheter in a patient who is septic, who remains hypotensive despite 20 mcg/kg per minute of dopamine, and who has started to exhibit a decrease in urine output. Improvisation in patient care requires considerable expertise; it is the antithesis of following a protocol.

Experts do not use guidelines or analytical reasoning when faced with familiar problems. Instead, they generate a single course of action—they do what normally works. An expert sees the world differently than does a novice. An expert spots things that other people miss: how things usually work, patterns, anomalies, fine details but at the same time the big picture, and soft underbellies for attack. The expert is not using tacit knowledge as a substitute for the explicit analytical method, but as an improvement over it.¹⁶²

Consider the case of a patient who is admitted to the ICU with fever, respiratory distress, and hypotension. The resident examines the patient, but is unable to find an obvious source for the fever. Chest radiography and urinalysis are negative. The resident makes a presumptive diagnosis of septic shock. He begins therapy with vasopressors, fluids, and broad-spectrum antibiotics. The patient remains hypotensive. The next day, the attending physician hears a crunching sound over the apex, and suspects a Hamman crunch. At first, the physician is not convinced it is present, because the crunch is obscured by the noise of the ventilator. On closer listening, the crunch is synchronous with the heartbeat. The attending physician infers that the patient has an esophageal perforation, which is causing the sepsis. Surgical repair is successfully undertaken. The resident asks the attending physician, “How did you know what to look for?” “I don’t know why,” the physician replies, “I just sensed it.” This is an example of tacit knowledge used by the attending physician. The attending answers, “I don’t know why,” because the physician is unable to recall the precise train of thought that inspired the physician to suspect mediastinal emphysema. In part, the attending physician’s thinking is based on extensive experience, and the many similar cases the physician has seen in the past—an example of casuistry. The physician’s inspired guess, diagnostic acumen, does not stem solely from tacit knowledge. It is also based on explicit knowledge. Hearing a crunching sound in the precordium is of no value without the book learning that tells the physician what the sound signifies. The case exemplifies the intermingling of tacit and explicit knowledge.

The skills of an outstanding diagnostician are like those of the connoisseur, a refined ability to discern between nuances governed by a heightened sensibility (good taste). It is a mistake to imagine that the listing of a series of explicit steps in a protocol can substitute for the skills of a connoisseur.

Clinical Judgment

To understand the contribution of the above five forms of reasoning, it is useful to consider each on its own. Expert clinicians, however, call on all these forms of reasoning, and blend them together without using each on an independent basis. The amalgam of these forms of reasoning constitutes *clinical judgment*—a term widely used but rarely explicated.

Clinical judgment refers to the totality of mental resources and cognitive processes that a physician uses in deciding what to do for a particular patient. A good judge is someone who examines different pieces of evidence; he or she compares critically and impartially the merits of each, and makes a decision as to where the truth lies. An experienced clinician blends tacit and explicit knowledge in the process of making clinical decisions. Such blending of knowledge cannot be reduced to, or captured in, a mathematical model that would apply universally in all critically ill patients. It is obviously impossible to control or standardize tacit knowledge and clinical judgment among the clinicians participating in a randomized controlled trial. The marked variation in degree of skill among participating physicians is an important unmeasurable factor that can confound the outcome of a trial. Recognition of this factor has led to calls for the inclusion of an extra arm in a clinical trial that reflects the usual care of the participating physicians.

Conflation of Therapy and Monitoring

The fundamental goal of monitoring is to guide a clinician in management decisions. Thus, monitoring is inextricably linked to treatment. A monitoring device on its own never cured anyone. It can improve outcome only if it leads to more effective therapy. This link is not always recognized. Yet it hovers over all assessments of a monitoring device. The conflation of monitoring with therapy differs from the evaluation of diagnostic techniques in other areas of medicine; a test is judged by its diagnostic accuracy, not on the expectation that it will improve outcome.

Consider the use of hemodynamic monitoring in an unstable patient with septic shock. Therapeutic options include fluids, vasopressors, or inotropic agents. None of these agents directly correct the underlying problem. Each may also cause harm. When a supranormal cardiac index was used as a target for hemodynamic management in 762 critically ill patients, survival was no better than with usual care.¹⁵⁷ When high-dose dobutamine was used to achieve a high cardiac index, Hayes et al¹⁶⁴ found that treated patients had a higher mortality than the usual care group: 54% versus 34%. In a complex patient with septic shock, when initial hemodynamic therapy is unsuccessful, it is common to undertake invasive monitoring (often involving a PA catheter) to guide more aggressive therapy. An unsuccessful outcome in such patients may be the result of ineffective pharmacologic agents or complications caused by them. Yet, the results of such trials are commonly interpreted as evidence that hemodynamic monitoring is not effective.

Patient Response to Therapy

In medicine, we treat patients, not diseases. Patients have varying responses to therapy depending on their disease. Although the right therapy may be chosen (e.g., cefotaxime for meningococemia), a patient may have an undesired response to the therapy that worsens outcome. A PA catheter may be inserted in a

patient in septic shock to guide therapy. The patient may develop cardiogenic pulmonary edema (with fluid resuscitation), a life-threatening arrhythmia (with dobutamine), or digital ischemia (with norepinephrine). These adverse reactions can outweigh beneficial effects of the monitoring device (the PA catheter) itself. No monitoring modality is likely to improve outcome unless it is combined with a therapeutic intervention that has a good benefit-to-risk ratio.

Final Outcome

An outcome event is an occurrence (such as death) that may be modified after implementing some intervention (monitoring wedge pressure with a PA catheter). Outcomes research is defined by the statistical analysis of clinical data to determine whether particular interventions are associated with particular results. To discriminate between benefits and risks of an intervention, an adequate sample size is necessary. Sample size is calculated on the basis of baseline event rate, expected benefit, level of significance (α), and the power to detect a difference ($1-\beta$).¹⁶⁵ Studies evaluating the effect of monitoring on patient outcome often lack statistical power. Statistical power is especially important when complications are rare and the expected impact (of an intervention) is small. If the outcome under evaluation occurs frequently, fewer patients will be needed than if the outcome occurs infrequently. Moller et al^{129,166} conducted a prospective, randomized trial of the effect of pulse oximetry on the rate of postoperative complications in 20,802 surgical patients. Hypoxemia (defined as a $SpO_2 < 90\%$) was detected nineteen times more often in the oximeter group than in the control group ($p < 0.0001$). The primary goal of the study, however, was to determine whether the rate of postoperative complications was decreased. For this goal, statistical significance was not reached. The investigators concluded that it would take at least 500,000 patients to show a reduction in a rare event such as myocardial infarction and 1,900,000 patients to show a reduction in anesthesia-related deaths.

The American Society of Anesthesiologists introduced monitoring standards that mandated use of pulse oximetry in every patient undergoing anesthesia. Subsequently, Eichhorn¹⁶⁷ observed a threefold reduction in the incidence of major preventable intraoperative injury (ten in 757,000 vs. one in 244,000 anesthetics), yet the reduction did not reach statistical significance. Because the incidence of serious adverse events is low, it is estimated that an additional 6,545,605 cases would be needed to detect a significant reduction in major preventable intraoperative injury.^{24,168} Given the necessary effort, would anyone suggest that it makes sense to undertake such a study to find out if pulse oximetry is helpful during anesthesia? If the study were negative, would anesthesiologists stop using this monitoring device? Unlikely.

When anesthesiologists were surveyed in the study of Moller et al,^{129,166} 80% felt more secure when they used a pulse oximeter. Of 104 anesthesiologists, nineteen believed that the pulse oximeter helped them avoid serious complications (such as esophageal intubation, tracheal tube disconnection or displacement, anesthesia machine failure, and respiratory problems immediately after extubation). It is this mindset that has established pulse oximetry as an essential component of the standard of care despite a failure to achieve a $p < 0.05$ level of proven efficacy. The lack of concurrence between proof of efficacy in randomized trials of pulse oximetry and its requirement as part of the standard of care has major ramifications for all monitoring techniques.

When researchers evaluate the usefulness of a monitoring device, the selected outcome measure (mortality or complication rates) may be unrealistic. If benefit is expected to be small, an outcome based on mortality may not be sufficiently sensitive to detect it. Undertaking inappropriately designed studies may cause valuable tools to be discarded. At face value, the outcome of the study of Moller et al¹²⁹ should lead physicians to stop using pulse oximetry—at least in the operating room.

Authors commonly cite examples where intermediate outcomes (prevention of hypertension, hypoxemia, or ischemia; or suppressing premature ventricular contractions following myocardial infarction) were misleading. A widely cited study is the Cardiac Arrhythmia Suppression Trial (CAST) study. This consisted of a randomized trial to test the hypothesis that suppression of asymptomatic or mildly symptomatic premature ventricular contractions (PVCs) with antiarrhythmic agents in patients who have suffered a myocardial infarction improves survival.¹⁶⁹⁻¹⁷² It was believed that suppression of PVCs would prevent ventricular tachycardia and ventricular fibrillation and thus decrease mortality. Patients treated with antiarrhythmic agents (flecainide, encainide, or moricizine), however, had a higher mortality (8.0%, sixty-three of 755) than patients receiving placebo (3.5%, twenty-six of 743).

The increased mortality in the intervention arm of the CAST study is often interpreted as evidence that physiologic reasoning is unreliable, and that mortality is the only study outcome that reliably captures truth of efficacy.⁴⁹ Before accepting that conclusion, the data need to be closely scrutinized. A number of concerns arise. Mortality in the placebo group (3.5%) was much lower than expected.¹⁷³ The low mortality was partly a consequence of study design.^{174,175} Only patients whose ventricular arrhythmias were suppressed during an open-label titration phase of the study were randomized to the active drug or placebo. Patients whose ventricular arrhythmias were not suppressed or who did not tolerate the pharmacologic agent were not randomized to either arm.^{169,172} Thus, patients randomized into the study had a low risk for death for two reasons: The exclusion of patients who had died and exclusion of patients in whom arrhythmias could not be suppressed (with drugs) during the open-label titration (prerandomization) phase.¹⁷⁵ Of patients in the placebo arm, 80% had PVCs alone without having nonsustained ventricular tachycardia; moreover, more than half had a near-normal ejection fraction (above 40%).¹⁷² There is further evidence that mortality in the placebo group is not generalizable. The CAST investigators undertook a subsequent analysis in the 318 patients who survived the open-label titration phase but who were not randomized to either arm of the study and the 942 patients who were randomized to the placebo arm.¹⁷⁶ Occurrence of death or a resuscitated cardiac arrest was higher in the nonrandomized patients (8.5%, twenty-seven of 318) than in the placebo group (4%, thirty-eight of 942).¹⁷⁶ In summary, the adverse effects of the antiarrhythmic agents in the CAST study may largely reflect the failure of the investigators to recruit representative patients into the placebo arm, insofar as the investigators excluded patients in whom they could not suppress their

arrhythmias (a marker of increased mortality).^{171,177–179} Certainly, it is inappropriate to use data from the CAST study to conclude that therapy guided by ECG monitoring (that is, physiologic reasoning) is not helpful.

Conclusion

Patients receiving mechanical ventilation are vulnerable to a host of complications and it is essential to spot these problems at the earliest possible time. Such complications are commonly associated with alterations in physiologic functions in different organ systems. By continuously recording many physiologic variables in real time, monitoring systems offer the possibility of alerting a physician to a significant change in a patient's status. The hope is that such a warning will provide an opportunity to interrupt a cascade of events sufficiently early in their evolution to avert a lethal catastrophe. Accordingly, expertise in the operation of monitoring systems plays a vital part in the intelligent use of mechanical ventilation. This expertise involves more than familiarity with the touch pads and screens of the machinery. Physicians must know the physiologic principles on which each monitor is based, appreciate the responsiveness and fidelity of the charted variables, be facile in troubleshooting a system, and evaluate monitored data in accordance with the canons developed for interpreting diagnostic tests. All these skills play a vital part in minimizing the ills that the ventilated patient is heir to—they form a consummation devoutly to be wished.

Acknowledgments

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